

Anesthesia and Perioperative Medicine

Ashley Matthews Shilling

The term *athlete*, derived from the Greek word *athlon* meaning “prize” or “contest,” conjures images of a healthy, fit person. Although athletes possess physical speed, strength, and talents, they also have a repertoire of medical problems as well as a number of sports-related injuries leading them to the operating room. This chapter discusses common medical problems in the sports patient within the context of surgical and anesthetic implications. Perioperative athletic concerns are also discussed, including preparation of the sports medicine patient for surgery, anesthetic choices, pain management, and common postoperative complications and concerns.

PREOPERATIVE CARE AND CONCERNS: COMMON MEDICAL ISSUES FOUND IN ATHLETES

Respiratory Considerations

Asthma

A common pulmonary disorder in athletes, asthma is marked by restriction of airflow in the lungs, narrowing of the airways, and inflammation of the bronchi. Bronchoconstriction and inflammation induce shortness of breath, wheezing, coughing, and chest tightness. The prevalence of exercise-induced bronchospasm is 8% to 12% in the general population but may be as high as 50% in elite athletes and those competing in cold-weather sports.¹ Several high-profile athletes with asthma, including Billy Jean King, Dennis Rodman, Jackie Joyner-Kersey, and Amy Van Dyken, have performed at the highest level with adequate treatment.

Triggers include smoking, environmental, respiratory viruses, odors, and exercise. The specifics of diagnosis and therapy for persons with asthma are detailed in a separate chapter in this book. Asthma can be diagnosed with a thorough history and physical examination and various provocative tests. Although multiple options are available to treat asthma, an important tenet is prevention by avoidance of triggering agents and maintenance therapy. Common strategies for maintenance and treatment of exacerbations include β_2 agonists (with albuterol being the most commonly used), inhaled corticosteroids, systemic corticosteroids, anticholinergic drugs, leukotriene-receptor antagonists, and methylxanthine agents. Drugs may be administered by nebulizer, metered-dose inhaler, orally, or intravenously in emergency situations. Before a patient undergoes elective surgery, it is important that asthma be optimally controlled. Despite the bronchodilating

properties of most anesthetics, asthmatics are more prone to bronchospasm after airway manipulation under anesthesia. Avoidance of general anesthesia and endotracheal intubation may be beneficial. When a general anesthetic is needed, a laryngeal mask airway has been shown to cause less airway reactivity than an endotracheal tube.²

Obstructive Sleep Apnea

Obstructive sleep apnea (OSA) is defined by periods of hypoventilation or apnea caused by airway obstruction during sleep. The severity of disease can be determined by the number of obstructive episodes, degree of oxygen desaturation, number of microarousal episodes, and number of arrhythmias per unit of time. Up to 90% of individuals with moderate to severe sleep apnea may remain undiagnosed.³⁻⁵

OSA may be more common in athletes than originally thought.⁵ A study of eight randomly selected NFL teams and 302 players demonstrated a 14% incidence of sleep apnea with a 34% incidence in linemen.⁶ Dr. Archie Roberts, a retired heart surgeon who has teamed with the Mayo Clinic to screen retired NFL players, has noted an alarming incidence of OSA. The incidence of sleep-disordered breathing was present in 52.3% of the former football players.⁷ Furthermore, a study of sumo wrestlers in Tokyo found that 11 of 23 wrestlers had significant sleep-disordered breathing.⁸ Clearly, OSA is a problem that entails clinical vigilance, even in athletes who are presumed to be healthy.

Obstruction of the upper airway during sleep or sedation occurs from a decrease in oropharyngeal and laryngopharyngeal tone and leads to partial or complete airway obstruction.⁹ An increase in airway resistance, hypoxia, and hypercarbia ensue, resulting in an increase in sympathetic tone and subsequent patient arousal. Complications of OSA include chronic hypoxia, hypoxemia, and hypercarbia leading to pulmonary hypertension, right heart failure, and sudden death. Less obvious associations between OSA and long-term sequelae include depression, hypertension, coronary artery disease, and stroke.

OSA is a significant problem in the postoperative period. Sedatives, narcotic drugs, hypnotic agents, and muscle relaxants can exacerbate symptoms. Because of pain and the inherent stress of surgery, patients experience disturbances in rapid eye movement sleep patterns for several days after the immediate postoperative period. The increased risk of obstruction and other complications should be considered when planning intraoperative

Abstract

The perioperative period for patients undergoing sports medicine procedures is critical in improving outcomes. Patients should be optimized preoperatively. Pain management and ideal anesthetic choices depend on surgical and patient factors and should involve multi-modal analgesia techniques including regional procedures. Peripheral nerve blocks are ideal for the sports medicine patient, and the use of ultrasound has made these procedures safe and effective in the perioperative period.

Keywords

neuraxial anesthesia
peripheral nerve block
local anesthetic systemic toxicity (LAST)
multi-modal analgesia
American Society of Anesthesiologists (ASA) sedation definitions

and toxicologic data suggested that 134 of these deaths were related to cardiac pathology, of which 36% were the result of HCM. Physical examination in asymptomatic athletes who have occult HCM may be normal or may reveal a fourth heart sound and/or a left ventricular lift (nonspecific). In persons with a significant obstruction, a harsh crescendo-decrescendo systolic murmur may be heard that begins shortly after S1, is typically best auscultated at the apex and lower left sternal border, and may radiate to the axilla and base, yet not usually into the neck. As a result of increased obstruction, the murmur may increase with an assumption of an upright posture from a supine, sitting, or squatting position or after the Valsalva maneuver.

It is beyond the scope of this chapter to detail various diagnostic and therapeutic tools or to discuss the controversies of screening athletes for HCM. Pre-participation screening for HCM is an ardently debated subject, with conflicting recommendations from varying organizations. The American Heart Association has indicated that a thorough pre-participation history and physical examination may be adequate to detect underlying pathology in many cases and that more detailed testing (to improve sensitivity), including an electrocardiogram and transthoracic echocardiogram, be reserved for cases with features that raise concern on clinical examination.¹⁶ Conversely, in Italy, pre-participation screening for athletes includes a detailed patient and family history, physical examination, and electrocardiogram (with additional cardiac testing as needed).¹⁷ After institution of the mandated nationwide systematic screening program, the incidence of sudden cardiovascular death in young competitive athletes between 1979 and 2004 significantly decreased in the Veneto region of Italy. Therapeutic interventions included medications (β - or calcium channel blockers and antiarrhythmic agents), pacemakers and/or defibrillators, and/or invasive, surgical management (i.e., a myomectomy or ablation of the cardiac septum).

Diabetes Mellitus

Diabetes mellitus (DM) is prevalent in athletes. Although athletes of different ages may have either type 1 or type 2 DM, type 1 DM occurs more commonly in athletes, and it is estimated that about 215 000 people in the United States who are younger than 20 years have some form of diabetes.¹⁸ Long-term complications of diabetes include cardiovascular disease, cerebrovascular disease, peripheral neuropathy, retinopathy, and nephropathy. Diagnostic tests include fasting plasma glucose level, oral glucose tolerance, or random plasma glucose level tests. Fasting glucose levels should be less than 100 mg/dL; levels between 100 and 125 mg/dL reflect impaired glucose tolerance, and levels greater than 200 mg/dL raise concern for diabetes.

Treatment of diabetes consists of maintaining normal or near-normal glucose levels and preventing hypoglycemia. Although hypoglycemia rarely occurs in persons with type 2 DM, it can lead to significant morbidity and mortality in persons with type 1 DM. Severe hypoglycemia can lead to seizures and irreversible brain damage due to the brain's need of glucose for normal metabolic activity. Hypoglycemia should be immediately treated by administration of glucose in the form of carbohydrates, glucose tablets, or IV dextrose. In the perioperative period, hypoglycemia

may be missed in sedated or anesthetized patients, and hyperglycemia may lead to an increased number of wound infections and poor wound healing.

INTRAOPERATIVE CARE AND CONSIDERATIONS OF THE SPORTS MEDICINE PATIENT

Orthopaedic Surgeries and Pain

A study of ambulatory surgeries showed that orthopedic patients have the highest incidence of postoperative pain.¹⁹ Inadequately treated pain can significantly affect recovery time, rehabilitation, patient satisfaction, and may increase readmission rates. When choosing the anesthetic, various factors should be considered, including the precise location of surgical intervention, extent of bone manipulation, instrumentation, osteotomies, ligamentous involvement, expected or planned nerve manipulation, use of bone or tendon autografts, size of incisions, the use of intraoperative tourniquets, and the requirements for postoperative physical therapy and range-of-motion (ROM) exercises. These perioperative factors will help the health care providers predict the level of postoperative pain, determine if a regional technique is indicated, and determine the needed duration of effect.

A peripheral nerve block (PNB) may provide excellent analgesia postoperatively, but may not be sufficient to cover the entire surgical field or for intraoperative retraction and manipulation. The necessary positioning for surgery can induce non-operative discomfort and can be a limiting factor in using a regional technique as the primary anesthetic. Pain induced by use of a tourniquet can be a source of substantial discomfort for a patient who is awake. Occlusion times more than 45 to 60 minutes may lead to break-through ischemic pain and may be intolerable despite the use of a PNB. Finally, certain surgical procedures put the patient at high risk for nerve injury or compartment syndrome and the surgeon may wish to identify this risk early in the postoperative course. In this setting, a regional technique may confound the postoperative examination for the first 24 hours or may mask a compartment syndrome.

Patient Characteristics

Several clinical features, including the aforementioned medical comorbidities, play an important role in shaping the anesthetic plan. In the younger athletic population, cardiovascular risk factors are fortunately rare, as GA is known to increase the risk of significant cardiac events in persons with certain cardiac risk factors.²⁰ For persons with OSA, the use of non-narcotic means of analgesia and avoidance of GA is recommended to reduce the risk of perioperative complications.²¹⁻²⁵ If a patient has a history of difficult intubation, it may be beneficial to avoid airway instrumentation to prevent potential airway complications. Patients with increased risk of aspiration as a result of preexisting esophageal or gastric pathology, or more commonly as a result of gastroesophageal reflux disease, may also benefit from the avoidance of GA. Patient psychological factors may play a prominent role in the anesthetic choice. Promising young athletes may fear a protracted nerve injury after undergoing a regional technique.

Regional Anesthesia

The term regional anesthesia implies analgesia and immobility of a region of the body. Regional anesthesia techniques may be further subdivided into those that target segmental spinal levels (i.e., central or neuraxial blocks such as spinal or epidural blocks), peripheral nerves (e.g., PNBs and plexus blocks), or extremities via IV regional block (i.e., Bier block). Neuraxial blocks and PNBs are achieved with the use of local anesthetic agents, narcotics, and other adjuvant medication. Field blocks typically describe the more superficial injection of local anesthetics. Both neuraxial nerve blocks and PNBs can be modified by the insertion of a catheter to continuously infuse medication during surgery, as well as for postoperative analgesia. Peripheral nerve catheters can be used in an inpatient or outpatient setting with proper patient selection and education. PNBs and neuraxial anesthetic agents may be used as the primary anesthetic or as an adjuvant to GA.

Contraindications to Regional Anesthesia

Absolute and relative contraindications to regional anesthesia typically vary by the type of planned intervention and the patient. Patient refusal, the inability to lie still during administration of a block, infection at the site of injection, recent use of thrombolytic agents, and operator inexperience for the proposed regional technique are perhaps the only shared absolute contraindications. Increased intracranial pressure may lead to brainstem herniation, which precludes use of neuraxial procedures. Other relative contraindications include evidence of systemic infection, preexisting neurologic disease, coagulopathy attributable to intrinsic disease or medication, or aberrant anatomy. All risks and benefits should be weighed thoroughly before proceeding with regional anesthesia.

Complications of Regional Anesthesia

The incidence of infection associated with neuraxial techniques is approximately 1 in 40,000.^{26,27} The risk of PNB infection has yet to be clearly defined, but appears to be greater with the use of continuous, indwelling catheters in the axillary or femoral areas for more than 48 hours.^{28,29} Skin contaminants either by colonization or local infection, systemic infection via seeding of catheter sites, or immune suppression enhance risk.³⁰ Spinal hematomas may form after induction of spinal or epidural anesthesia in patients with underlying coagulopathy.³¹ Detailed recommendations for management are well described in the American Society of Regional Anesthesia 2010 Practice Advisory.³² Although the data are less instructive for PNBs, the literature includes rare case reports of a hematoma causing nerve injury, which seems to be more prevalent with use of deeper techniques such as lumbar plexus block. Nerve injuries are a major concern for both the orthopedic surgeon and anesthesiologist, yet the overall risk of peripheral nerve injury (PNI) is fortunately small. A meta-analysis in 2007 showed rates of permanent nerve injury between 0 and 7.6 per 10,000 for neuraxial procedures and approximately 1 in 15,000 for PNBs.³³ Recent studies of PNI demonstrate long-term neurologic findings widely ranging from 0.029% to 0.2%, largely due to study methodology.³⁴⁻⁴⁰ Most

commonly, the incidence of neurologic findings after PNB are quoted between 2 and 4/10,000.⁴¹ The incidence of short-term paresthesias or dysesthesias is thought to be much higher at 1% to 3%, with the highest incidence following interscalene brachial plexus blocks.³³

The American Society of Regional Anesthesia (ASRA) published a practice advisory in 2016 with a specific focus on neurologic injury following regional techniques. In addition to reviewing the incidence, causes, and risks for neurologic injury following regional techniques, the advisory also reviews risks of PNI after common orthopedic procedures and describes treatment algorithms following PNI.⁴¹

Benefits of Regional Anesthesia in the Sports Medicine Patient

The benefits of regional anesthesia are abundant. Compared with narcotic analgesic agents, postoperative pain control is notably improved with peripheral nerve and neuraxial blocks.⁴²⁻⁴⁵ As expected, these studies also demonstrate less nausea, vomiting, ileus, pruritus, and sedation because of the opiate-sparing effects. It is also well established that the use of regional anesthesia improves functional recovery in select patients. Peripheral nerve catheters resulted in earlier ambulation, shorter hospital stays, and shorter rehabilitation periods after discharge, as well as improved ROM and mobilization after knee surgery.⁴⁶⁻⁴⁸ Epidural catheters also offer similar analgesic benefits when compared with peripheral nerve catheters, although hypotension and nausea are more common.⁴⁹ In the ambulatory setting, orthopedic procedures can account for a significant percentage of unplanned readmissions for pain control.⁵⁰ Use of outpatient perineural catheters resulted in fewer unplanned admissions after discharge in patients undergoing anterior cruciate ligament (ACL) surgery.⁵¹

Regional anesthesia may also benefit patients in measures that transcend pain control and its associated secondary gains. Operative blood loss may be decreased through decreased arterial and venous pressure and blockade of sympathetic tone in the affected extremity.⁴⁶ The incidence of deep venous thrombosis (DVT) and pulmonary embolism (PE) may be lower with neuraxial anesthesia because of increased blood flow, decreased viscosity, and modification of inflammatory mediators.^{52,53} Neuraxial anesthesia may also reduce the risk of infections at surgical sites, possibly through modulation of inflammation and immune responses.⁵⁴ Furthermore, the avoidance of GA in high-risk patient populations is an immeasurable benefit.

Tourniquets

A precious commodity during orthopedic procedures, tourniquets provide a bloodless field for surgery. However, complications associated with tourniquet placement are not uncommon. Pain with resulting hypertension and tachycardia, "sickling" in sickle cell disease, nerve injury, and tissue damage may result.⁵⁵ Nerve injury can result from ischemia, pressure, or shearing at the edges of the tourniquet. The pain is typically dull and begins with inflation. Progressive symptoms may include tingling, numbness, paresthesia, and a change in quality of pain. Pain typically begins at 20 minutes and can become severe at approximately

40 minutes.⁵⁶ No duration of tourniquet use is guaranteed to be safe, yet most experts consider 2 hours to be the “maximum ischemic time.”⁵⁷ At the conclusion of procedures, release of the tourniquet can lead to hypotension and metabolic acidosis as metabolites from an ischemic limb are released into the circulation.⁵⁸ In patients with congestive heart failure, a sudden increase in circulating volume can lead to volume overload and acute heart failure. Occluding venous blood flow distal to the tourniquet predisposes a patient to thrombi formation. Systemic release of the thrombi after tourniquet deflation may lead to stroke and PE.⁵⁸

Positioning

Although many orthopedic surgeries may be performed in a supine intraoperative position, some operations require more complicated positions and padding. Proper positioning and padding is essential to minimize complications.

Beach Chair Position

The beach chair position has become the chosen intraoperative position for many orthopedic surgeons who perform procedures on the shoulder. This position is favored for many reasons, including better exposure of the shoulder joint, avoidance of the lateral position, decreased risk for brachial plexus injuries, and reduced bleeding in the joint. Nonetheless, the beach chair position has multiple physiologic consequences, including a decrease in mean arterial blood pressure, central venous pressure, stroke volume, cardiac output, and arterial oxygen content. Additional concerns include venous pooling, increased risk for air embolus, and poor access to the airway. Reports of fluid extravasation from the shoulder joint into soft tissue have led to airway compromise.

An important concern is the discrepancy that can exist between cerebral perfusion and the site of blood pressure measurement. This issue is especially vital in the hypertensive patient who may have a rightward shift of the cerebral autoregulatory curve and needs higher mean arterial blood pressure to maintain adequate perfusion. Blood pressure should be measured in the arm at the brachial artery, and the distance between the heart and external auditory meatus should be calculated. The hydrostatic gradient can be determined using the conversion of 0.77 mm Hg decrease for every centimeter height difference between the brain and the site of blood pressure measurement.⁵⁹ Additionally, placing a blood pressure cuff on the ankle (which could be as much as 150 cm below the brain) has been implicated in providing the anesthesiologist and surgeon with a false sense of increased cerebral perfusion pressures. Serious neurologic injury, presumably due to inadequate cerebral perfusion during beach chair positioning, has been documented in four patients.⁶⁰

SPECIFIC REGIONAL ANESTHESIA TECHNIQUES

Neuraxial Anesthesia

Epidural, caudal, and spinal blocks are collectively referred to as neuraxial anesthetics. Spinal block also may be referred to as an intrathecal or subarachnoid block. Neuraxial techniques can be used for surgical anesthesia for any lower extremity, hip, or

pelvic procedure. Most commonly, a mix of a local anesthetic (with or without a narcotic) is used, but other adjuncts exist to enhance block quality and duration. Placement of a catheter, which is most commonly used with epidural blockade, facilitates the ability to augment or lengthen the duration of the block intraoperatively or provide ongoing postoperative analgesia. The chief difference between epidural and spinal blockade is the location of drug administration. Whereas a spinal block requires puncture of the dura mater and injection of the drug into the cerebrospinal fluid, an epidural block requires injection of drug past the ligamentum flavum, outside the dura mater. Because of varying anatomy within the epidural space and the presence of vasculature, fat, and potential septations, an epidural block can occasionally result in a patchy or unilateral block quality when compared with a spinal block.

The most common adverse effects encountered with neuraxial blocks are hypotension, pruritus, nausea, sedation, and urinary retention. Hemodynamic effects, including hypotension and bradycardia, are more noticeable with a spinal block, which is due to rapid onset of sympathectomy, as well as blockade of cardioaccelerator fibers in the upper thoracic spinal levels. Pruritus and nausea are more pronounced with use of a narcotic in the neuraxial mixture. Postdural puncture headache is another complication of neuraxial anesthesia. Believed to occur from an alteration in cerebrospinal fluid dynamics due to dural puncture, the incidence is fortunately low (0% to 2%) with spinal anesthesia when performed by an experienced provider with a small-gauge, noncutting needle.⁶¹⁻⁶³ Typically, an epidural block should not interrupt the dura mater, yet an accidental puncture can occur. A “wet tap” or cerebrospinal fluid return during epidural placement with a large 17-gauge Tuohy needle can induce a headache in a significant number of patients. In such cases, definitive treatment with an epidural blood patch is highly effective. The use of certain local anesthetic agents and traumatic needle placement may increase this risk. Major complications such as “high spinal,” local anesthetic toxicity, infection, neurologically significant bleeding, and permanent nerve damage are extremely rare.

Practically, for outpatient sports medicine procedures, neuraxial blocks can be used to limit risks and costs associated with GA and subsequent recovery. In the setting of outpatient procedures such as knee arthroscopy, comparisons between GA and epidural, spinal, or PNBs have yielded conflicting results regarding speed, cost, and complications.⁶⁴⁻⁶⁸ Although discharge readiness may be expedited for neuraxial anesthetics when considering pain control, mental status, and nausea, other factors, such as urinary retention and time for spinal to regress, may ultimately delay discharge.

Peripheral Nerve Blocks

Overview

Surgical anesthesia, immobility, and analgesia can be achieved with the precise deposition of local anesthetics and adjuvants near peripheral nerves. This can be accomplished by using surface landmarks, nerve stimulation, and ultrasound-guided techniques. The duration of the block may be manipulated by altering the type of local anesthetic or adjuvant or by placement of a perineural catheter. Localization of the nerves to be blocked may

involve several techniques. Incorporating knowledge of the surface anatomy along with palpation of vessels and bony structures can help identify the location for blockade. PNBs were initially performed by provoking paresthesias in the desired nerve distribution with needle advancement. For nerve stimulation, an insulated needle is used to apply a low-level current at the tip, which can elicit a twitch in a corresponding muscle group supplied by a target nerve. Alternatively, “real-time” ultrasonography may be used to visualize the peripheral nerves and needle, which enables anesthesiologists to administer medications with great anatomic specificity. Thus far, no one technique has been shown to decrease neurologic complications.⁴¹ There is mounting evidence that ultrasound can decrease local anesthetic toxicity by allowing visualization of the spread of local anesthetics and also by decreasing the total amount of local anesthetic needed to provide an adequate block.^{37,69-72} Ultrasound-guided blocks have numerous advantages, including a faster onset of action, improved quality, decreased volume of local anesthetic used, and a decrease in accidental vascular puncture rates.⁷²⁻⁷⁴

Upper Extremity Innervation and the Brachial Plexus

The brachial plexus forms from the ventral rami of spinal nerves C5 to T1, with occasional contribution from C4 and T2. Fig. 29.2 shows the brachial plexus. The plexus exits between the anterior and middle scalene muscles of the neck, courses over the first rib and under the clavicle, surrounds the axillary artery, and divides into peripheral nerves that supply motor and sensory innervation to the upper extremity. A fascial sheath surrounds the plexus from the intervertebral foramina to the upper arm and can serve both to compartmentalize injected medication and provide a safe barrier to this approach. Several techniques can be used to block the brachial plexus, each of which differs in coverage of the plexus and adverse effect profiles. Fig. 29.2 demonstrates the different approaches used to block the brachial plexus.

Shoulder and Upper Arm Procedures

Interscalene Block

An interscalene block provides excellent anesthesia to the upper brachial plexus C5-C7, including the entire glenohumeral joint, rotator cuff, the lateral two thirds of the clavicle, and the proximal humerus. It frequently spares blockage of C8-T1, thus often does not provide block to the medial brachial cutaneous nerve and ulnar hand distribution. Interscalene block is therefore most useful for shoulder, upper arm, and forearm procedures. It is possible to perform shoulder surgery with an interscalene block and avoid GA. Many surgeons prefer this approach because it facilitates continuous neurologic assessment for the patient in the beach chair position. As previously mentioned, an anesthetized patient in this position can experience hypotension, leading to cardiovascular and cerebrovascular complications. The interscalene block is the most commonly performed regional technique for shoulder surgery since it blocks both the axillary and suprascapular nerves. A 2015 meta-analysis showed improved pain control up to six hours during movement and eight hours at rest as well as a reduction in opioid use up to 24 hours postoperatively in patients who received an interscalene

PNB.⁷⁵ Continuous catheter techniques have been shown to extend these benefits with improvements in sleep and patient satisfaction as well.⁷⁵

The interscalene block is performed in the anterolateral neck in close proximity to the exiting nerve roots adjacent to the scalene muscles (Fig. 29.3). The proximity to the cervical plexus, the phrenic and recurrent laryngeal nerves, stellate ganglion, and other nerves of the head and neck can lead to side effects of the block. Blockade of the ipsilateral phrenic nerve almost always occurs, so this block is relatively contraindicated for persons with severe lung disease or a contralateral diaphragmatic paralysis. However, in the vast majority of healthy patients, a phrenic nerve block causes few or no symptoms and seldom requires treatment other than reassurance. Hoarseness and Horner syndrome are also possible, but usually pose only a minor inconvenience. Fortunately, all these effects typically resolve as the block wanes. More serious complications, such as vertebral artery injection, high spinal, and accidental spinal cord injection, have been documented but are rare.

Suprascapular Block

A suprascapular block can be used alone or in conjunction with an axillary nerve block for patients with contraindications to the interscalene block and phrenic nerve paresis.⁷⁶ The nerve supplies the muscles of the rotator cuff (except for the subscapularis), the glenoid, most of the joint capsule, the posterior scapula, part of the acromion and acromioclavicular joint, as well as the skin of the neck and most of the superior shoulder. The nerve is typically blocked via a posterior approach from the suprascapular notch either with nerve stimulation or ultrasound. The risk of inducing a pneumothorax is very low with this technique. A study comparing the interscalene, supraclavicular, and suprascapular block in patients undergoing shoulder surgery demonstrated the suprascapular block to have significantly less effect in mean vital capacity reduction. There was no significant difference in mean pain scores and 24-hour opioid consumption for the three different block approaches.⁷⁷

Elbow, Forearm, and Hand Procedures

Supraclavicular Block

Performed at the level of the trunks and divisions of the brachial plexus in the supraclavicular fossa, the supraclavicular block provides coverage throughout most of the arm. It spares the upper shoulder proximally and the T2 distribution of the medial arm. The intercostobrachial nerve, a cutaneous nerve derived from the T2 intercostal nerve, provides sensation to the medial upper arm and is typically spared by brachial plexus approaches. Therefore a subcutaneous band of local anesthetic high in the axilla is used to block this nerve when an upper arm tourniquet is used.⁷⁸ Historically the supraclavicular block was avoided by many practitioners who feared a pneumothorax or puncture of the subclavian artery. However, the use of ultrasound and the ability to visualize the first rib and pleura have made this block one of the most popular upper extremity PNBs. The block evolves rapidly because of the dense concentration of nerves at this level. Insertion of a perineural catheter with this approach has been described for prolonged analgesia.⁷⁹ The supraclavicular block

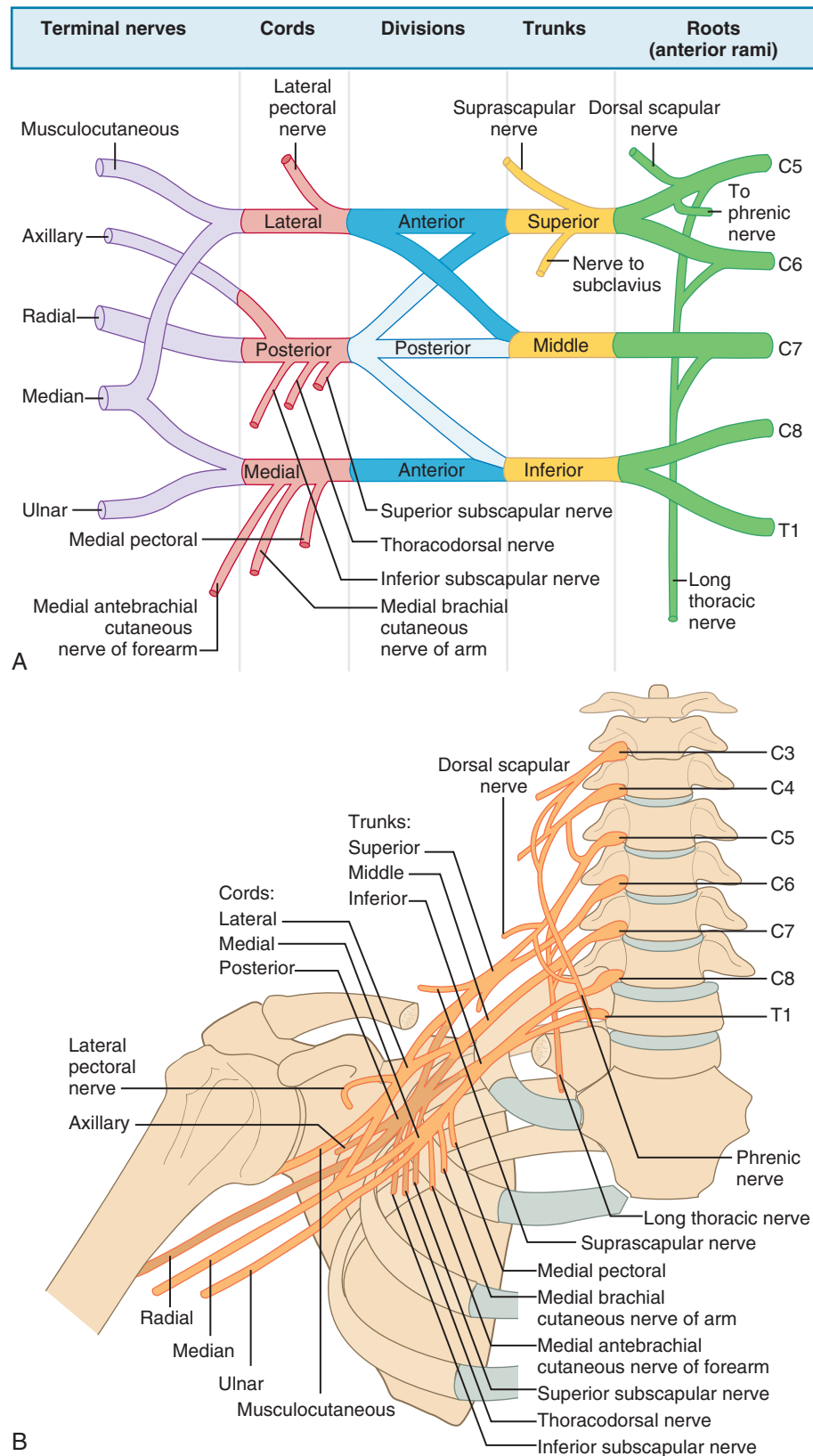


Fig. 29.2 (A) A schematic representation of the brachial plexus. (B) The brachial plexus in situ. (From Bogart BI, Ort VH. *Elsevier's Integrated Anatomy and Embryology*. Philadelphia: Mosby; 2007.)

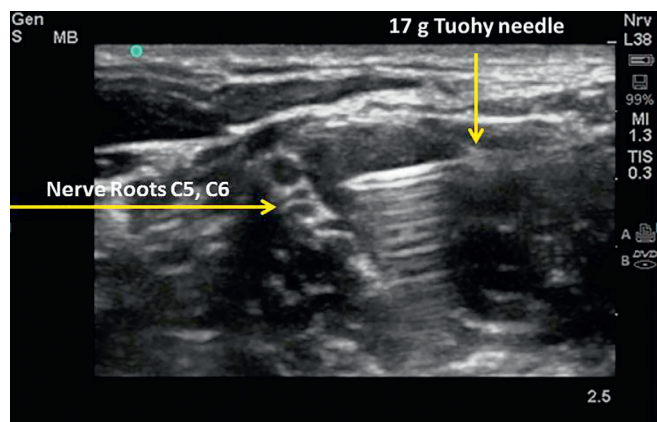


Fig. 29.3 An ultrasound image of the interscalene approach to the brachial plexus.

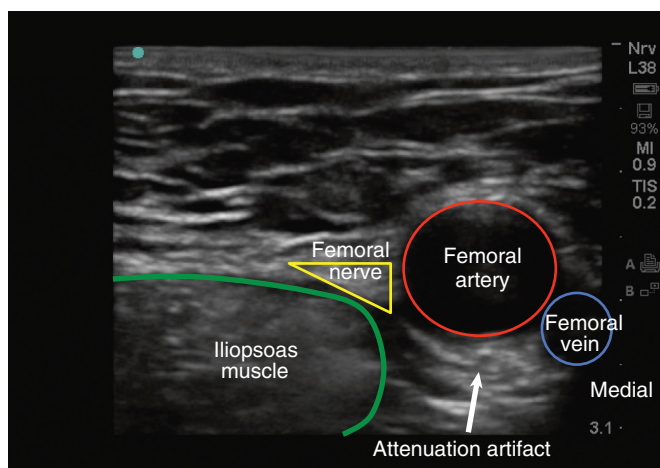


Fig. 29.4 An ultrasound image of the supraclavicular approach to the brachial plexus.

has potential adverse effects akin to the interscalene block, including phrenic nerve involvement (40% to 60%).⁸⁰

Fig. 29.4 shows ultrasound images of the supraclavicular approach to the brachial plexus.

Infraclavicular Block

The infraclavicular block provides coverage similar to the supraclavicular block with a much lower risk of pneumothorax and phrenic nerve involvement, and therefore is more suitable for persons with lung disease. At this level, the deeper location of the nerves beneath the pectoralis muscles can make needle placement more challenging. Ultrasound visualization may be difficult in obese or very muscular patients. The block is performed inferior to the clavicle and slightly medial to the coracoid process where the lateral, posterior, and medial cords of the plexus surround the axillary artery. Speed of block onset is slower than the supraclavicular approach, but this approach is ideal for a continuous catheter due to the deeper target nerves with more tissue in which the catheter can anchor.

Axillary Brachial Plexus Block

The axillary block targets the plexus as it divides into the terminal peripheral nerves. Contrary to its name, the axillary nerve was

often spared prior to the use of ultrasound because the nerve usually divides from the posterior cord of the brachial plexus more cephalad. The axillary block provides excellent coverage for forearm and hand procedures. This approach has fewer adverse effects and complications compared to interscalene or supraclavicular blocks because it is performed outside of the thorax, avoiding the lungs and numerous nerves and vessels of the neck. Additionally, this block is ideal for patients with coagulopathies because it is superficial and blood vessels may be compressed against the humerus. Many approaches and techniques for this block have been described, including use of simple landmarks and palpation of the axillary artery, nerve stimulation, and ultrasound.

Lower Extremity Innervation

The lower extremity peripheral nerves are formed by the lumbar and sacral plexuses and are supplied by anterior rami of spinal nerves L1-L4 and L4-S3, respectively. For complete anesthesia at these levels, many practitioners would choose a neuraxial block rather than plexus blocks because of its simplicity in covering multiple nerves with one procedure. In addition to other terminal peripheral branches, the lumbar plexus gives rise to the lateral femoral cutaneous, femoral, and obturator nerves, whereas the sacral plexus forms the sciatic nerve and the posterior femoral cutaneous nerve. As a result, many of these lower extremity peripheral nerves can be blocked outside the pelvis. Because of the complexity of innervation, complete anesthesia for many procedures requires combination blocks.

Hip, Femur, Anterior Thigh, and Knee Procedures

Femoral Nerve Block

The hip and ligaments of the knee are common sites of surgery in the orthopedic sports population. While ligament repairs of the knee have been performed as an outpatient procedure for decades, hip arthroscopies are newer to the outpatient arena and lend their own challenges. Pain can be significant and studies demonstrate postoperative pain scores ranging from seven to ten on the 10-point visual analogue scale (VAS).^{81,82} Unfortunately, due to the complex innervation of the hip and involvement of both the lumbar and sacral plexus, performing regional anesthetic techniques can be challenging.

Several small studies have examined intra-articular local anesthetic and portal local anesthetic injection with some evidence to suggest benefit in pain scores, though data is limited by small study size.⁸³⁻⁸⁵ A number of regional techniques have been performed in hip arthroscopy patients including femoral nerve blocks, fascia iliaca blocks (with target nerves of the femoral and lateral femoral cutaneous nerves), and lumbar plexus blocks. Insufficient data exists to perform a meta-analysis, but there is evidence that femoral nerve blocks and lumbar plexus blocks offer pain benefits when compared to intravenous pain medications.^{86,87} There is less compelling evidence to show benefits of fascia iliaca blocks.⁸⁶

The femoral nerve block is common in outpatient surgery and is used most frequently for knee procedures, though more recently the adductor canal has largely replaced the femoral nerve block due to favorable motor-sparing benefits. A femoral

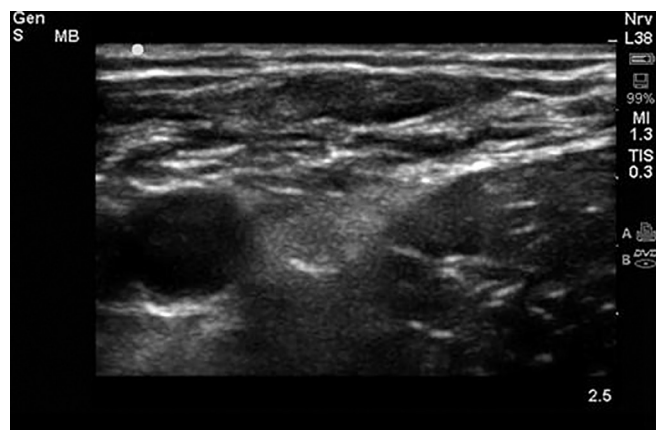


Fig. 29.5 An ultrasound image of the femoral nerve, artery, and vein.

nerve block can be performed with relative ease using landmark, nerve stimulation, and ultrasound techniques because of the relatively shallow location of the nerve distal to the inguinal ligament and the predictable location relative to the femoral pulse. Blockade results in anesthesia of the anterior thigh and knee, quadriceps, anterior joint capsule, and the medial calf (via the saphenous nerve). ACL reconstruction surgery is one of the more commonly performed orthopedic surgeries and a procedure that may lead to significant pain. Intra-articular injections have not been shown to be as effective as regional techniques.^{88,89} Data show that patients receiving femoral nerve blocks have decreased pain scores, decreased opioid consumption, decreased PACU stay, and decreased risks for unplanned admissions, with the addition of the sciatic block adding additional benefits over a femoral block alone.^{90,91}

Catheter placement is simple, and significant improvements in postoperative pain have been described when a femoral block is used for outpatient ACL repair.⁹² The risks of the femoral block are minimal, but may include intravascular injection, hematoma, and infection (after use of a catheter). **Fig. 29.5** demonstrates an ultrasound examination of the femoral nerve.

Adductor Canal Nerve Block

The adductor canal block has largely replaced the femoral nerve block for procedures of the knee. Due to blockade of the quadriceps muscles with a femoral nerve block, many anesthesiologists are now performing a more distal block in hopes of sparing motor effects but still providing sensory block. The adductor canal houses the saphenous nerve, nerve to the vastus medialis, and branches of the anterior obturator nerve. A meta-analysis comparing adductor canal block versus femoral canal block for total knee arthroplasty showed no significant differences in pain scores in patients receiving adductor canal versus femoral nerve blocks and also demonstrated that the adductor canal block had significant improvement of quadriceps strength over the femoral nerve block.⁹³ Limitations of this meta-analysis are that only six studies and 408 patients were included and that the studies showed variability in how they defined the site of an adductor canal block. In the orthopedic sports medicine population, the adductor canal block provided noninferior analgesia and superior quadriceps strength compared with femoral nerve block in ACL

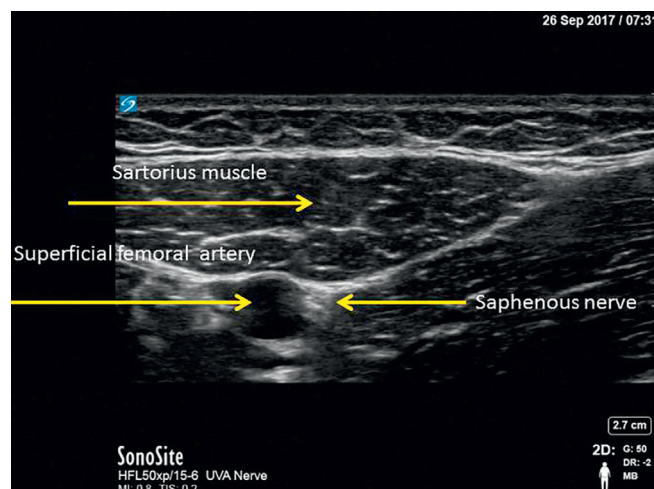


Fig. 29.6 An ultrasound image of the adductor canal block.

reconstruction.⁹⁴ While reports of falls due to motor weakness are rare, motor-sparing techniques facilitate early ambulation and may decrease falls in at-risk patients. Additionally, an injury incurred from the block is less likely to involve motor weakness as a complication. **Fig. 29.6** shows an ultrasound image of the adductor canal block.

Lower Leg, Ankle, and Foot Procedures

Sciatic Nerve Block

There are a number of approaches to the sciatic nerve, beginning at the sciatic notch and extending down to the popliteal fossa where it divides into the tibial and common peroneal nerves. Nerve stimulation or ultrasound guidance is usually required for the deeper approaches. Analgesia of the leg below the knee can be completely achieved with a combination of sciatic plus saphenous nerve block. Blockade of the large sciatic nerve is more involved and takes longer to establish but may last longer than other PNBs. Catheters may be inserted via popliteal or subgluteal approaches with equal analgesia for foot surgery.⁹⁵ Studies combining sciatic with femoral catheters for knee arthroplasty have shown improved analgesia. However, no clear benefit is seen on functional measures, and early postoperative rehabilitation may be impaired by weakness when a sciatic nerve block is added to the femoral.⁹⁶

Saphenous Nerve

The selective saphenous nerve block is used in conjunction with a popliteal block for complete anesthesia below the knee. It can be performed either above the knee in the adductor canal as noted above or below the knee using a field block. The former approach is associated with a higher degree of success.⁹⁷

Ankle Block

The five terminal branches of the sciatic and femoral nerves can be individually blocked at the ankle to provide anesthesia and analgesia for foot procedures, provided no tourniquet is used above the ankle. If bone surgery is planned above the midfoot, this technique may be inadequate and a popliteal/saphenous approach should be used. The block can be executed entirely with

landmarks alone, yet latency to onset is prolonged and patient discomfort is greater because of the need for multiple injections. Ultrasound has enhanced this block in clinical practice.

Intravenous Regional Anesthesia (Bier Block)

The Bier block is one of the older methods of providing regional anesthesia. Intravenous injection of a local anesthetic with a tourniquet creates a high venous pressure gradient that encourages the flow of the anesthetic to the smaller nerve endings of the extremity. The Bier block is best suited for short procedures that last less than 60 minutes, because pain from the tourniquet will develop. This block is most frequently used for upper extremity procedures, but it may be used for lower extremity surgery. A larger volume of medication is necessary for a lower limb block, which increases the risk for the most common complication of this block—local anesthetic toxicity. The risk of toxicity can be minimized by ensuring excellent tourniquet occlusion for a minimal time interval. With a lidocaine Bier block, 45 minutes is recommended for a sufficient amount of the drug to bind to tissues, which ensures safe serum levels prior to cuff release.⁹⁸ Soft tissue procedures, tendon repairs, and minor joint repairs are all acceptable applications.

Continuous Catheter Techniques

It is estimated that 50% to 70% of surgeries are being performed in the ambulatory setting; thus more complex sports and orthopedic procedures are occurring on an outpatient basis.⁹⁹ Total shoulder arthroplasty, unicondylar and total knee replacements, and multi-ligament knee surgeries are now standard procedures in ambulatory surgical centers. Management of pain with continuous catheter techniques has become standard of practice. Continuous brachial plexus catheters for shoulder, elbow and hand surgeries, popliteal catheters for foot and ankle surgeries, and femoral or adductor catheters for knee procedures have been shown to benefit patients and improve numerous outcomes. In addition to prolonging analgesic effects of a nerve block, lower doses of local anesthetics can be used, which can result in less motor block as well as a decrease in diaphragmatic paresis.¹⁰⁰

Continuous nerve catheters have been shown to improve pain scores, decrease narcotic needs, and improve sleep in patients undergoing orthopedic procedures.⁴³ Additionally, outpatient continuous nerve catheters are safe if appropriate patient screening and education is performed.¹⁰¹ Clinicians must carefully select appropriate catheter patients with attention to patient resources and understanding of the management of a continuous catheter. Relative exclusions for sending a patient home with a continuous nerve catheter include: patient lacks understanding or comfort managing or removing catheter; patient without caregiver present to monitor for signs of toxicity; patients without a means to communicate any problems or concerns; and patient lives far from medical facility to receive help if needed.

Local Anesthetic Agents

Classified into two chemical groups (amides and esters), local anesthetic agents impede transmission of nerve impulses by blocking voltage-gated sodium channels. Amide local anesthetic drugs include bupivacaine, ropivacaine, lidocaine, and mepivacaine.

Ester local anesthetic agents include procaine, chlorprocaine, tetracaine, and benzocaine. Amide local anesthetic drugs are degraded in the liver, and ester local anesthetic drugs are degraded by enzymes in the plasma. Compared with the amide group, ester anesthetic agents are more likely to induce allergic reactions as a result of para-aminobenzoic acid (PABA), a metabolite of ester anesthetics.¹⁰²

Local anesthetic drugs differ in potency, speed of onset, duration of action, and toxicity.¹⁰³ The most commonly used local anesthetic drugs in clinical practice include bupivacaine, ropivacaine, lidocaine, and mepivacaine. Bupivacaine is the longest acting and most potent drug, but it has the highest degree of cardiotoxicity of any anesthetic agent, and thus a maximum dose of 2.5 mg/kg is typically recommended. Ropivacaine is less potent than bupivacaine and is less cardiotoxic; the maximum recommended dose is 3 mg/kg. Lidocaine, although less potent and shorter acting than either bupivacaine or ropivacaine, has a very stable toxicity profile. The maximum dose is 4.5 mg/kg without epinephrine and 7 mg/kg with added epinephrine.¹⁰⁴ Mepivacaine has a rapid onset of action and a longer duration than lidocaine, but it has a shorter duration than either bupivacaine or ropivacaine. Mepivacaine is a very useful anesthetic for regional blocks for patients who desire a primary regional anesthetic but do not want a prolonged nerve block postoperatively. The maximum dose is 300 mg without epinephrine and 500 mg with epinephrine. Adding epinephrine to a local anesthetic at a concentration of 1:200 000 increases the duration of analgesia by approximately 200%.¹⁰⁵ The addition of more concentrated epinephrine does not provide added duration benefits and can induce tachycardia and hypertension. Other adjuvants used in local anesthetic agents that can help improve the quality and duration of PNBs include clonidine, dexamethasone, dexmedetomidine, and buprenorphine.

Local Anesthetic Systemic Toxicity

Adverse events from local anesthetic drugs range from mild symptoms to severe neurologic and cardiovascular symptoms caused by systemic absorption and unintentional intravascular injection. Symptoms often begin with tinnitus, a metallic taste, dizziness, and lightheadedness and rapidly progress to seizures and ultimately to cardiovascular collapse.¹⁰⁶ Seizures, agitation, and loss of consciousness are the most common signs of neurologic symptoms. Central nervous system (CNS) symptoms can be treated with benzodiazepines such as midazolam, lorazepam, or diazepam. In addition, managing the airway becomes imperative because hypoventilation and respiratory acidosis can prolong and worsen the effects of local anesthetic toxicity.¹⁰⁷

In high doses, local anesthetic drugs delay cardiac conduction and depress contractility, and bupivacaine has a particularly high affinity for cardiac sodium channels.^{108,109} Randomized controlled trials to study local anesthetic systemic toxicity are needed, and although local anesthetic systemic toxicity is a rare event, it was traditionally thought to occur in up to 1 in 1000 PNBs.¹¹⁰ From retrospective studies, it is known that half of all cases of toxicity are seen within 50 seconds of injecting a local anesthetic but that signs can be observed up to 60 minutes after the injection.¹¹⁰

Arrhythmias are the most common cardiac signs of toxicity. However, asystole can occur in 12% of patients. These findings do not apply if a local anesthetic is injected directly into the vertebral arteries or the carotid artery, after which a seizure can occur with a small dose of a local anesthetic.¹¹⁰

Treatment depends on the extent of the symptoms. Minor CNS effects do not require intervention and will typically resolve if no additional local anesthetic is injected. For major reactions, including CNS symptoms and cardiovascular collapse, lipid emulsion therapy has shown promise as a “first-line” treatment. Marketed under the trademark name Intralipid, lipid emulsion therapy is thought to function as a “lipid sink” that extracts hydrophobic drugs, such as bupivacaine, from cardiac tissue. Additionally, it is thought that Intralipid (Baxter Healthcare, Deerfield, IL) may stabilize myocardium and act as an energy substrate to cardiac myocytes.

It is administered intravenously as a bolus of 1.5 mL/kg, followed by an infusion of 0.25 mL/kg/minute, and is continued for 10 minutes after hemodynamic stabilization.¹¹⁰ The nearest medical facility with the capability for cardiopulmonary bypass should be notified in case of complete cardiopulmonary collapse.¹¹¹ Algorithms have been published that guide the clinician through treating a patient with LAST. Table 29.1 shows the treatment of LAST.

PERIOPERATIVE PHARMACOLOGY

Perioperative Pain Management

Effective pain management can be challenging, particularly in the ambulatory setting, where patients must be discharged home with safe and effective medications. Ineffective pain control is a major cause of unanticipated hospital admissions after outpatient surgery. Narcotics have been the mainstay of acute postoperative pain control, yet they are associated with adverse effects, some of which can be lethal in a subset of patients (i.e., persons with OSA or significant respiratory disease). This section reviews the various options for perioperative analgesia. Prescription and nonprescription opioids have gained significant worldwide attention, as the opioid epidemic is now noted to be the worst drug crisis in the history of our country, with 91 Americans dying every day from opioid overdose.¹¹² While initially thought to be low risk, more recent studies demonstrate that patients prescribed short-term opioids following surgery are at increased risk of chronic opioid use as compared to patients who have never used opioids.¹¹³ Currently, there is little data on optimal doses of narcotics after orthopedic-specific procedures, though more studies are being performed to examine best practice and help guide clinicians on prescribing parameters.^{113–115,133}

The most commonly overdosed prescription drugs include methadone, oxycodone, and hydrocodone.¹¹² The American Pain Society released clinical practice guidelines for postsurgical pain management in February of 2016 in *The Journal of Pain*.¹¹⁶ These guidelines included 32 recommendations for optimal pain management in surgical patients. Table 29.2 lists the recommendations. STRONG, high-quality EVIDENCE recommendations were made for four interventions. Three of the four *strong recommendations with high quality of evidence* to support

TABLE 29.1 Treatment of Local Anesthetic Systemic Toxicity Checklist for Treatment of Local Anesthetic Systemic Toxicity

The pharmacologic treatment of local anesthetic systemic toxicity (LAST) is different from other cardiac arrest scenarios

- Get help
- Initial focus:
 - Airway management: ventilate with 100% oxygen
 - Seizure suppression: benzodiazepines are preferred; AVOID propofol in patients having signs of cardiovascular instability
 - Alert the nearest facility having cardiopulmonary bypass capability
- Management of cardiac arrhythmias:
 - Basic and advanced cardiac life support (ACLS) will require adjustment of medications and perhaps prolonged effort
 - AVOID vasopressin, calcium channel blockers, beta blockers, or local anesthetic
 - REDUCE individual epinephrine doses to <1 µg/kg
- Lipid emulsion (20%) therapy (values in parentheses are for 70 kg patient):
 - Bolus 1.5 mL/kg (lean body mass) intravenously over 1 minute (~100 mL)
 - Continuous infusion 0.25 mL/kg/min (~18 mL/min; adjust by roller clamp)
 - Repeat bolus once or twice for persistent cardiovascular collapse
 - Double the infusion rate to 0.5 mL/kg/min if blood pressure remains low
 - Continue infusion for at least 10 minutes after attaining circulatory stability
 - Recommended upper limit: Approximately 10 mL/kg lipid emulsion over the first 30 minutes
- Post LAST events at www.lipidrescue.org and report use of lipid to www.lipidregistry.org

From Neal JM, Bernards CM, Butterworth JF, et al. ASRA practice advisory on local anesthetic systemic toxicity. *Reg Anesth Pain Med*. 2010;35(2):151–161.

that are pertinent to orthopedic sports patients include the use of multi-modal analgesics, the use of acetaminophen, with or without non-steroidal antiinflammatory drugs (NSAIDs), and the use of regional techniques as a consideration. The recommendations stress increased awareness of multi-modal analgesia and the treatment of pain by targeting various receptors of pain modulation in addition to opioid receptors. Attention to Enhanced Recovery After Surgery (ERAS) protocols has prompted a number of perioperative interventions that occur before surgery and continue throughout the days and weeks after surgery. ERAS began in the 1990s by Professor Henrik Kehlet with the goal of providing evidence-based, multi-modal, multi-disciplinary perioperative care pathways designed for early recovery after surgical procedures.¹¹⁷ This begins before surgery with patient education and nutrition and continues throughout the perioperative period. Common components of ERAS protocols include close attention to multi-modal analgesia, fluid management, and early mobilization. Data is accumulating to show improved outcomes including earlier discharge, fast-tracking patients, decreased cost, and improved quality of care of surgical patients when ERAS protocols are implemented.

TABLE 29.2 Management of Postoperative Pain: A Clinical Practice Guideline From the American Pain Society, the American Society of Regional Anesthesia and Pain Medicine, and the American Society of Anesthesiologists' Committee on Regional Anesthesia, Executive Committee, and Administrative Council^a

Recommendation 1	The panel recommends that clinicians provide patient and family-centered, individually tailored education to the patient (and/or responsible caregiver), including information on treatment options for management of postoperative pain, and document the plan and goals for postoperative pain management (strong recommendation, low-quality evidence).
Recommendation 2	The panel recommends that the parents (or other adult caregivers) of children who undergo surgery receive instruction in developmentally appropriate methods for assessing pain as well as counseling on appropriate administration of analgesics and modalities (strong recommendation, low-quality evidence).
Recommendation 3	The panel recommends that clinicians conduct a preoperative evaluation including assessment of medical and psychiatric comorbidities, concomitant medications, history of chronic pain, substance abuse, and previous postoperative treatment regimens and responses, to guide the perioperative pain management plan (strong recommendation, low-quality evidence).
Recommendation 4	The panel recommends that clinicians adjust the pain management plan on the basis of adequacy of pain relief and presence of adverse events (strong recommendation, low-quality evidence).
Recommendation 5	The panel recommends that clinicians use a validated pain assessment tool to track responses to postoperative pain treatments and adjust treatment plans accordingly (strong recommendation, low-quality evidence).
Recommendation 6	The panel recommends that clinicians offer multi-modal analgesia, or the use of a variety of analgesic medications and techniques combined with nonpharmacologic interventions, for the treatment of postoperative pain in children and adults (strong recommendation, high-quality evidence).
Recommendation 7	The panel recommends that clinicians consider TENS as an adjunct to other postoperative pain treatments (weak recommendation, moderate-quality evidence).
Recommendation 8	The panel can neither recommend nor discourage acupuncture, massage, or cold therapy as adjuncts to other postoperative pain treatments (insufficient evidence).
Recommendation 9	The panel recommends that clinicians consider the use of cognitive-behavioral modalities in adults as part of a multi-modal approach (weak recommendation, moderate-quality evidence).
Recommendation 10	The panel recommends oral over IV administration of opioids for postoperative analgesia in patients who can use the oral route (strong recommendation, moderate quality evidence).
Recommendation 11	The panel recommends that clinicians avoid using the intramuscular route for the administration of analgesics for management of postoperative pain (strong recommendation, moderate-quality evidence).
Recommendation 12	The panel recommends that IV PCA be used for postoperative systemic analgesia when the parenteral route is needed (strong recommendation, moderate-quality evidence).
Recommendation 13	The panel recommends against routine basal infusion of opioids with IV PCA in opioid-naïve adults (strong recommendation, moderate-quality evidence).
Recommendation 14	The panel recommends that clinicians provide appropriate monitoring of sedation, respiratory status, and other adverse events in patients who receive systemic opioids for postoperative analgesia (strong recommendation, low-quality evidence).
Recommendation 15	The panel recommends that clinicians provide adults and children with acetaminophen and/or NSAIDs as part of multi-modal analgesia for management of postoperative pain in patients without contraindications (strong recommendation, high-quality evidence).
Recommendation 16	The panel recommends that clinicians consider giving a preoperative dose of oral celecoxib in adult patients without contraindications (strong recommendation, moderate-quality evidence).
Recommendation 17	The panel recommends that clinicians consider use of gabapentin or pregabalin as a component of multi-modal analgesia (strong recommendation, moderate-quality evidence).
Recommendation 18	The panel recommends that clinicians consider IV ketamine as a component of multi-modal analgesia in adults (weak recommendation, moderate-quality evidence).
Recommendation 19	The panel recommends that clinicians consider IV lidocaine infusions in adults who undergo open and laparoscopic abdominal surgery who do not have contraindications (weak recommendation, moderate-quality evidence).
Recommendation 20	The panel recommends that clinicians consider surgical site-specific local anesthetic infiltration for surgical procedures with evidence indicating efficacy (weak recommendation, moderate-quality evidence).
Recommendation 21	The panel recommends that clinicians use topical local anesthetics in combination with nerve blocks before circumcision (strong recommendation, moderate-quality evidence).
Recommendation 22	The panel does not recommend intrapleural analgesia with local anesthetics for pain control after thoracic surgery (strong recommendation, moderate-quality evidence).
Recommendation 23	The panel recommends that clinicians consider surgical site-specific peripheral regional anesthetic techniques in adults and children for procedures with evidence indicating efficacy (strong recommendation, high-quality evidence).
Recommendation 24	The panel recommends that clinicians use continuous, local anesthetic-based peripheral regional analgesic techniques when the need for analgesia is likely to exceed the duration of effect of a single injection (strong recommendation, moderate-quality evidence).
Recommendation 25	The panel recommends that clinicians consider the addition of clonidine as an adjuvant for prolongation of analgesia with a single-injection peripheral neural blockade (weak recommendation, moderate-quality evidence).
Recommendation 26	The panel recommends that clinicians offer neuraxial analgesia for major thoracic and abdominal procedures, particularly in patients at risk for cardiac complications, pulmonary complications, or prolonged ileus (strong recommendation, high-quality evidence).

TABLE 29.2 Management of Postoperative Pain: A Clinical Practice Guideline From the American Pain Society, the American Society of Regional Anesthesia and Pain Medicine, and the American Society of Anesthesiologists' Committee on Regional Anesthesia, Executive Committee, and Administrative Council—cont'd

Recommendation 27	The panel recommends that clinicians avoid the neuraxial administration of magnesium, benzodiazepines, neostigmine, tramadol, and ketamine in the treatment of postoperative pain (strong recommendation, moderate-quality evidence).
Recommendation 28	The panel recommends that clinicians provide appropriate monitoring of patients who have received neuraxial interventions for perioperative analgesia (strong recommendation, low-quality evidence).
Recommendation 29	The panel recommends that facilities in which surgery is performed have an organizational structure in place to develop and refine policies and processes for safe and effective delivery of postoperative pain control (strong recommendation, low-quality evidence).
Recommendation 30	The panel recommends that facilities in which surgery is performed provide clinicians with access to consultation with a pain specialist for patients with inadequately controlled postoperative pain or at high risk of inadequately controlled postoperative pain (e.g., opioid-tolerant, history of substance abuse) (strong recommendation, low-quality evidence).
Recommendation 31	The panel recommends that facilities in which neuraxial analgesia and continuous peripheral blocks are performed have policies and procedures to support their safe delivery and trained individuals to manage these procedures (strong recommendation, low-quality evidence).
Recommendation 32	The panel recommends that clinicians provide education to all patients (adult and children) and primary caregivers on the pain treatment plan, including tapering of analgesics after hospital discharge (strong recommendation, low-quality evidence).

^aRecommendations based on systematic review.

IV, Intravenous; NSAIDs, nonsteroidal anti-inflammatory drugs; PCA, patient-controlled analgesia; TENS, transcutaneous electrical nerve stimulation.

Opioids

Opioids, which are classified as natural (i.e., morphine) or synthetic (e.g., fentanyl, sufentanil, and remifentanil), have been the cornerstone of perioperative analgesia. However, unfavorable adverse effects include ventilatory depression, sedation, PONV, ileus, pruritus, and difficulty voiding. These problems can lead to unforeseen hospital admissions in the ambulatory surgery setting. Notwithstanding, opioids are the most commonly used perioperative pain medication. A significant advantage of opioids in the ambulatory setting is availability in parenteral form for use during procedures and in oral preparations for use at home.

The most commonly used IV opioids are morphine, hydromorphone, and fentanyl. Morphine, often considered the “gold standard opioid,” has a duration of action of 4 to 5 hours, is metabolized by the liver, and is cleared by the kidneys. Because it is converted by the liver into an active metabolite, it must be used with caution in patients with renal failure or insufficiency. Hydromorphone is five times as potent as morphine. Analgesia is achieved within 5 minutes, and compared with morphine, its duration of action is shorter.¹¹⁸ It is a better choice in patients with renal failure. Fentanyl is 100 times more potent than morphine, and onset of analgesia is within 3 to 7 minutes. Because of its short duration of action, fentanyl is a drug that must be administered more frequently. Additionally, it is safe to use in patients with renal failure.

Many commonly prescribed oral opioids are used in combination with acetaminophen, including a combination of hydrocodone and acetaminophen. Hydrocodone has a half-life of 2.5 to 4 hours and is thought to be a prodrug that requires the enzyme CYP2D6 to process and achieve analgesia. Patients lacking this enzyme may not have the expected pain relief from hydrocodone. Oxycodone is not a prodrug; it has a half-life of 2.5 to 4 hours, and it provides consistent pain relief.

Methadone

When caring for patients with chronic pain, a formal pain consultation can help guide pain management. Drugs such as methadone and buprenorphine can make acute surgical pain management in chronic pain patients extremely difficult. Methadone is a mu-receptor agonist and N-methyl-D-aspartate receptor antagonist that was developed in the 1930s as an alternative to addictive opioids. Methadone is used primarily for analgesia and for treatment from opioid addiction. Because of its antagonistic effects on NMDA receptors, it is a good drug for neuropathic pain. Other benefits include the longest half-life of opioids and low cost. Methadone can be administered orally, sublingually, or parenterally. Side effects include those of other opioids as well as prolonged QT and the potential for arrhythmias. Methadone has been used in the perioperative period in a number of orthopedic patients, most commonly spinal fusions.¹¹⁹⁻¹²² Care must be taken when caring for patients on methadone and determining how to manage additional analgesics safely.

Buprenorphine

Buprenorphine was made available in the United States in the 1970s, with the initial use for outpatient opioid detoxification and addiction therapy. Buprenorphine is an agonist at the mu receptor and an antagonist at the kappa and delta receptors. Several different variants have been approved, including oral tablets, sublingual tablets, buccal films, and skin implants. Several contain the opioid antagonist naloxone to help prevent intravenous abuse of the drug. Commonly used brands include Suboxone (buprenorphine/naloxone sublingual tablet) and Subutex (buprenorphine sublingual tablet). Buprenorphine is a semi-synthetic opioid that is classified as a partial opioid agonist. Benefits of this drug include a decreased potential for abuse and fewer characteristics of euphoria, thus less potential for dependence. While the effects of buprenorphine increase

with increasing doses, this effect reaches a plateau, thus called the ceiling effect. Other safety benefits include less respiratory depression as well as a reduction of opioid cravings. While advantages of this drug are clear, in the acute pain phase, patients on buprenorphine provide challenges. Patients on buprenorphine are resistant to traditional opioid analgesic effects and debates continue about whether patients should continue or stop buprenorphine prior to surgical procedures. Discontinuing 72 hours prior to surgery may be necessary with a bridge of oral analgesics.¹²³

Acetaminophen

Acetaminophen (Tylenol) is an excellent analgesic with minimal adverse effects at therapeutic doses. It works both centrally and peripherally by activating serotonergic, cholinergic, noradrenergic, cannabinoid, and N-methyl-D-aspartate receptors. It has few gastrointestinal adverse effects and little effect on platelet function. Hepatotoxicity is the most concerning adverse effect. Fortunately, it is extremely rare and is easily minimized by limiting a daily dose to 4 g.¹²⁴ Liver function is unaffected after therapeutic dosing in persons with stable liver disease. In patients with severe liver disease and chronic alcohol abuse, reducing the dose to 1 g three times a day is safe.¹²⁴ Typical dosing is 500 to 1000 mg every 6 hours. Acetaminophen can be administered orally, rectally, or intravenously with benefits to each of the different routes. While oral acetaminophen is susceptible to first-pass hepatic metabolism and absorption may be negatively affected by opioids, it is economically the best choice and can be administered preoperatively for peak effect prior to emergence from anesthesia. IV acetaminophen can be notably beneficial in patients who cannot tolerate oral administration. Marketed in the United States under the trademark name Ofirmev, therapeutic concentrations are achieved within 20 minutes and last for 2 hours.¹²⁴ The addition of IV acetaminophen in one study reduced postoperative morphine consumption by 20% in the first 24 hours.¹²⁵ Postoperative nausea and vomiting (PONV) was reduced by 30% and sedation by 29% in other studies, and data from 14 placebo-controlled randomized controlled trials with 1464 patients showed improved analgesia with IV acetaminophen.¹²⁵ Pain intensity scores over 24 hours were improved in patients undergoing joint replacement, as was time until rescue opioids, rescue medication consumption, and administration of opioids when acetaminophen was given.¹²⁶

Nonsteroidal Antiinflammatory Drugs

NSAIDs have antiinflammatory, antipyretic, and analgesic properties; they block the synthesis of the enzyme cyclooxygenase and slow the acute inflammatory response peripherally and hyperalgesia centrally. In persons undergoing ambulatory surgery, NSAIDs decrease opioid requirements and potentially reduce opioid-related adverse effects.¹²⁷ NSAIDs may be administered either orally or intravenously with similar benefit. NSAIDs have been shown to improve dynamic pain control. Selective and nonselective NSAIDs are used commonly with selective COX2 activity decreasing some of the side effects. Diclofenac and Celecoxib have been shown to be significantly more selective to COX2 with Ketorolac being nonselective. Despite this, Ketorolac is

frequently used by anesthesiologists in the perioperative period and can be administered intravenously, intramuscularly, or orally. Ketorolac reduces postoperative pain with minimal incidence of PONV and has been shown to decrease time to discharge.¹²⁸ The optimal dose is 15 to 30 mg every 6 hours for a maximum of 5 days. The concern exists that ketorolac may increase microvascular bleeding, worsen renal function, and impair bone healing, though studies in humans are inconclusive on the effects on bone healing. However, for most sports medicine procedures, NSAIDs such as ketorolac are excellent opioid-sparing adjuvants for pain relief.

Ketamine

Ketamine is a unique anesthetic drug with significant analgesic properties. It works as an antagonist at the N-methyl-D-aspartate receptor. At small doses, ketamine has analgesic properties that extend 24 to 48 hours into the postoperative period even though the plasma half-life is 17 minutes. This effect is due to reduction of “wind-up,” a hyper-excitability state of CNS sensitization. Preoperative administration of ketamine may reduce the requirement for narcotics by 40% to 60%.¹²⁹ Ketamine has been shown in multiple studies to reduce opioid tolerance and to reduce opioid-induced hyperalgesia. A single dose of intraoperative ketamine has a 50% morphine-sparing effect in the first 48 hours after surgery.¹³⁰ Disturbing hallucinations and acute psychosis-like symptoms have been described in patients receiving ketamine, though usually at doses higher than those used for analgesia.¹³¹ Consequently, midazolam may be co-administered if ketamine is administered in doses greater than 1 mg/kg in a patient who is awake. Ketamine can also lead to tachycardia and hypertension.

Propofol

Propofol was first discovered in the 1970s and has since become the mainstay of induction agents in anesthesia. It is also used for sedation, typically in the form of an infusion. Its site of action is primarily on the GABA A receptor and has the clinical benefit of being very short-acting due to rapid redistribution. A single bolus has effects lasting only minutes. Two major benefits of propofol are its amnestic effects and anti-emetic effects. Side effects include pain on injection, hypotension, and apnea. Propofol decreases cerebral blood flow and also intracranial pressure. Propofol is the drug commonly used for a total intravenous anesthetic (TIVA) in patients with malignant hyperthermia.

Dexmedetomidine

Dexmedetomidine was approved as a sedative in the United States in 1999 and offers unique properties. As an alpha-2 receptor agonist, its site of action on presynaptic neurons decreases norepinephrine release with inhibition of postsynaptic activation. The effect on the central nervous system, particularly the locus ceruleus, leads to the intended effect of sedation with preservation of patient cooperation. Unique to dexmedetomidine, the sedation resembles nonrapid eye movement (REM) sleep.¹³² Dexmedetomidine also provides a myriad of benefits, including analgesia without significant respiratory effects. Upper airway tone is preserved and respiratory drive is maintained, thus making

it a nice choice in patients with respiratory co-morbidities and OSA.¹³³ Initially used as a sedative in mechanically ventilated patients, dexmedetomidine has become widely used in the perioperative setting for sedation, and as an adjunct to general anesthesia to reduce opioid and other anesthetic requirements. Side effects of dexmedetomidine include, most commonly, hypotension and bradycardia.

Sedation in the Orthopedic Patient

When providing sedation to patients undergoing minimally invasive orthopedic procedures such as reducing a fracture or superficial procedures using local anesthetics, it is critical to understand the recommendations of the ASA for standards of care. The ASA has published guidelines that define levels of sedation and standards for these four various levels of sedation.¹³⁴ These were approved by the ASA House of Delegates in 1999 and most recently amended in October of 2014. The four different levels of anesthesia, which are a continuum and can change throughout the procedure, are defined in Table 29.3. The ASA Task Force provides a number of recommendations based on the level of sedation and these include (1) presedation patient evaluation; (2) preprocedure preparation; (3) monitoring; (4) record keeping; (5) availability of an individual who is dedicated to patient monitoring and safety; (6) education of clinician providing sedation; (7) availability of rescue medications, airway devices, and oxygen; (8) appropriate use of sedative medications; and (9) aspects of recovery.¹³⁴

TABLE 29.3 The American Society of Anesthesiologists Definitions of Sedation

Minimal sedation (anxiolysis) is a drug-induced state during which patients respond normally to verbal commands. Although cognitive function and physical coordination may be impaired, airway reflexes and ventilatory and cardiovascular functions are unaffected.

Moderate sedation/analgesia ("conscious sedation") is a drug-induced depression of consciousness during which patients respond purposefully^a to verbal commands, either alone or accompanied by light tactile stimulation. No interventions are required to maintain a patent airway, and spontaneous ventilation is adequate. Cardiovascular function is usually maintained.

Deep sedation/analgesia is a drug-induced depression of consciousness during which patient cannot be easily aroused but respond purposefully^a following repeated or painful stimulation. The ability to independently maintain ventilatory function may be impaired. Patients may require assistance in maintaining a patent airway, and spontaneous ventilation may be inadequate. Cardiovascular function is usually maintained.

General anesthesia is a drug-induced loss of consciousness during which patients are not arousable, even by painful stimulation. The ability to independently maintain ventilatory function is often impaired. Patients often require assistance in maintaining a patent airway, and positive pressure ventilation may be required because of depressed spontaneous ventilation or drug-induced depression of neuromuscular function. Cardiovascular function may be impaired.

^aReflex withdrawal from a painful stimulus is NOT considered a purposeful response.

From www.asahq.org/quality-and-practice-management/standards-guidelines-and-related-resources.

PERIOPERATIVE COMPLICATIONS IN THE SPORTS MEDICINE PATIENT

Allergic Reactions

Intraoperative anaphylaxis can occur in otherwise healthy patients. It results from the release of various chemical mediators of inflammation from basophils and mast cells with resultant dysfunction of skin, mucous membranes, gastrointestinal, cardiovascular, and pulmonary systems. Its accurate incidence in the United States is unknown but is estimated at approximately 1:10,000 to 1:20,000 patients.^{135,136} The most common agents include muscle relaxants, latex, and antibiotics. Less common causes include opioid drugs, local anesthetic agents, chlorhexidine, colloid drugs, and hypnotic agents such as propofol.¹³⁷ Very small quantities of the inciting substance are required to initiate the pathophysiologic cascade of anaphylaxis, and signs can be observed within a few seconds after the administration of drugs. Signs include hypotension, tachycardia, bronchospasm, angioedema, arrhythmias, cardiovascular collapse, and cardiac arrest. Immediate treatment is pivotal to sustaining life. The suspected drug should be discontinued promptly; the airway should be maintained with 100% oxygen supplementation; epinephrine, antihistamine drugs, and corticosteroid agents should be administered without delay; the patient should be placed in a Trendelenburg position; and surgery should be aborted until the patient is stabilized.

Parsonage-Turner Syndrome

Also known as *brachial plexus neuritis* or *neuralgic amyotrophy*, Parsonage-Turner syndrome (PTS) is a rare syndrome that can affect athletes in the perioperative period or may arise extraneous to a surgical event. Although it occurs infrequently (1.64 cases per 100,000 people), it is a disease process that has an atypical presentation and may be confusing to the orthopedic surgeon.¹³⁸ PTS was first described in 1897 by Feinburg, but it was not until the mid 1900s that Parsonage and Turner described more than 100 cases of this condition.^{139,140} The syndrome usually involves the proximal brachial plexus and can affect many lower motor neurons of the plexus. Many reports have been made of isolated nerve findings, with weakness of the anterior interosseous nerve being a common presentation. The syndrome is more prevalent in males than females and can affect persons of all ages; approximately one fourth of all cases of PTS follow a systemic illness, and patients typically present with pain in the brachial plexus (mostly in the shoulder region) followed by motor weakness.¹⁴¹

PTS has been linked to a number of inciting events. Etiologies include surgery, trauma, PNB, viral and bacterial infections, vaccinations, and systemic illnesses. PTS can occur after surgery and can be difficult to differentiate from a surgical or anesthesia-related injury. PTS is a diagnosis of exclusion and is mostly clinically based. Nerve conduction studies and electromyography may be helpful in locating the lesion and ruling out unrelated nerve compression or injury. Encouragingly, most patients recover spontaneously and require only supportive care. A review of 99 patients revealed that complete recovery from PTS has been noted in approximately 90% of patients, but that recovery may

be slow and can take up to 3 years. These data contradict another smaller study that showed that 50% of patients had residual, long-term deficits.¹⁴²

For a complete list of references, go to [ExpertConsult.com](https://www.expertconsult.com).

SELECTED READINGS

Citation:

Neal J. ASRA practice advisory on local anesthetic systemic toxicity. *Reg Anesth Pain Med*. 2010;35(2):151–161.

Level of Evidence:

II, systematic review of Level II studies

Summary:

Local anesthetic systemic toxicity (LAST) is a rare but potentially devastating clinical entity after use of local anesthetic agents. This practice advisory reviews the most current information known about the epidemiology, prevention, and treatment of LAST.

Citation:

Gersh BJ, Maron BJ, Bonow RO, et al. 2011 ACCF/AHA guidelines for the diagnosis and treatment of hypertrophic cardiomyopathy: executive summary: a report of the American College of Cardiology Foundation/American Heart Association task force on practice guidelines. *Circulation*. 2011;124:2761–2796.

Level of Evidence:

I, systematic review of Level I studies

Summary:

Hypertrophic obstructive cardiomyopathy is the most common cardiovascular cause of sudden cardiac death in young athletes. This excellent review outlines and grades the latest evidence and makes expert recommendations regarding management of patients with this complex condition.

Citation:

Antonakakis JG, Ting PH, Sites B. Ultrasound-guided regional anesthesia for peripheral nerve blocks: an evidence-based outcome review. *Anesthesiol Clin*. 2011;29(2):179–191.

Level of Evidence:

I, systematic review of Level I studies

Summary:

Regional anesthesia has become a mainstay of many orthopedic procedures, and interest in using ultrasound to provide regional anesthesia is explosive. This article provides an excellent review of current data, mostly consisting of randomized controlled trials, that compare ultrasound-guided nerve blocks with the traditional approach to nerve blocks.

Citation:

Neal JM, Barrington MJ, Brull R, et al. The second ASRA practice advisory on neurologic complications associated with regional anesthesia and pain medicine executive summary. *Reg Anesth Pain Med*. 2015;40:401–430.

Level of Evidence:

Ranges from II-IV

Summary:

This article defines the incidence of neurologic complications associated with peripheral nerve blocks, with a focus on the pathophysiology of mechanical, ischemic, and neurotoxic causes. Additionally, there is focus on neurologic complications associated with common orthopedic procedures. The advisory discusses modifiable risks for peripheral nerve injury. Additionally, the paper reviews the diagnosis and treatment of peripheral nerve injury and the level of evidence for these.

REFERENCES

1. Rundell KW, Jenkinson D. Exercise-induced bronchospasm in the elite athletes. *Sports Med.* 2002;32:583–600.
2. Brimacombe J. The advantage of the LMA over the tracheal tube or facemask: a meta-analysis. *Can J Anaesth.* 1995;42(11):1017–1023.
3. Chung F, Heinzer R, Vat S, et al. Prevalence of sleep-disordered breathing in the general population: the HypnoLaus study. *Lancet Respir Med.* 2015;3:310–318.
4. Chung F, Memtsoudis SG, Ramachandran SK, et al. Society of Anesthesia and sleep medicine guidelines on preoperative screening and assessment of adult patients with obstructive sleep apnea. *Anesth Analg.* 2016;123(2):452–473.
5. Chung F, Yegneswaran B, Liao P, et al. STOP questionnaire: a tool to screen patients for obstructive sleep apnea. *Anesthesiology.* 2012;108(5):768–775.
6. George C, Kab V, Levy A. Increased prevalence of sleep-disordered breathing among professional football players. *N Engl J Med.* 2003;348:367–368.
7. Albuquerqu F, Sert Kuniyoshi F, Calvin A, et al. Sleep-disordered breathing, hypertension and obesity in retired National Football League players. *J Am Coll Cardiol.* 2010;56(17):1432–1433.
8. Suzuki N, Fueta M, To M, et al. Sleep-disordered breathing among professional sumo wrestlers in Japan. International Conference of the American Thoracic Society, May 16–21. 2003.
9. Rama AN, Tekwani SH, Kushida CA. Site of obstruction in obstructive sleep apnea. *Chest.* 2002;122(4):1139–1147.
10. Chung F, Yegneswaran B, Liao P, et al. STOP questionnaire: a tool to screen patients for obstructive sleep apnea. *Anesthesiology.* 2008;108:812–821.
11. Ramachandran SK, Kheterpal S, Consens F, et al. Derivation and validation of a simple perioperative sleep apnea prediction score. *Anesth Analg.* 2010;110:1007–1015.
12. Topol EJ, Califf RM. *Textbook of Cardiovascular Medicine.* Vol. 355. Philadelphia: Lippincott Williams & Wilkins; 2007.
13. Maron BJ, Epstein SE, Roberts WC. Causes of sudden death in the competitive athlete. *J Am Coll Cardiol.* 1986;7:204–214.
14. Maron BJ, Gardin JM, Flack JM, et al. Prevalence of hypertrophic cardiomyopathy in a general population of young adults: echocardiographic analysis of 4111 subjects in the CARDIA study. *Circulation.* 1995;92:785–789.
15. Maron B, Shirani J, Poliac LC, et al. Sudden death in young competitive athletes; clinical, demographic, and pathological profiles. *JAMA.* 1996;276(3):199.
16. Gersh BJ, Maron BJ, Bonow RO, et al. 2011 ACCF/AHA guidelines for the diagnosis and treatment of hypertrophic cardiomyopathy: executive summary: a report of the American College of Cardiology Foundation/American Heart Association task force on practice guidelines. *Circulation.* 2011;124:2761–2796.
17. Corrado D, Basso C, Pavei A, et al. Trends in sudden cardiovascular death in young competitive athletes after implementation of a preparticipation screening program. *JAMA.* 2006;296(13):1593–1601.
18. National Diabetes Information Clearinghouse. National diabetes statistics. <http://diabetes.niddk.nih.gov/dm/pubs/statistics>. Accessed September 12, 2017.
19. Chung F, Ritchie E, Su J. Postoperative pain in ambulatory surgery. *Anesth Analg.* 1997;85(4):808–816.
20. Fleisher LA, Beckman JA, Brown KA, et al. ACC/AHA 2007 guidelines on perioperative cardiovascular evaluation and care for noncardiac surgery: executive summary: a report of the American College of Cardiology/American Heart Association Task Force on Practice Guidelines (Writing Committee to Revise the 2002 Guidelines on Perioperative Cardiovascular Evaluation for Noncardiac Surgery). *Anesth Analg.* 2008;106(3):685–712.
21. Boushra NN. Anaesthetic management of patients with sleep apnoea syndrome. *Can J Anaesth.* 1996;43:599–616.
22. Catley DM, Thornton C, Jordan C, et al. Pronounced, episodic oxygen desaturation in the postoperative period: its association with ventilatory pattern and analgesic regimen. *Anesthesiology.* 1985;63:20–63.
23. Gupta RM, Parvizi J, Hanssen AD, et al. Postoperative complications in patients with obstructive sleep apnea syndrome undergoing hip or knee replacement: a case-control study. *Mayo Clin Proc.* 2001;76:897–905.
24. Blake DW, Chia PH, Donnan G, et al. Preoperative assessment for obstructive sleep apnoea and the prediction of postoperative respiratory obstruction and hypoxaemia. *Anaesth Intensive Care.* 2008;36:379–384.
25. Gross JB, Bachenberg KL, Benumof JL, et al. Practice guidelines for the perioperative management of patients with obstructive sleep apnea: a report by the American Society of Anesthesiologists task force on perioperative management of patients with obstructive sleep apnea. *Anesthesiology.* 2006;104(5):1081–1093.
26. Moen V, Dahlgren N, Irestedt L. Severe neurological complications after central neuraxial blockades in Sweden 1990–1999. *Anesthesiology.* 2004;101(4):950–959.
27. Cook TM, Counsell D, Wildsmith JAW. Major complications of central neuraxial block: report on the Third National Audit Project of the Royal College of Anaesthetists. *Br J Anaesth.* 2009;102:179–190.
28. Capdevila X, Pirat P, Bringuier S, et al. French study group on continuous peripheral nerve blocks: continuous peripheral nerve blocks in hospital wards after orthopedic surgery: a multicenter prospective analysis of the quality of postoperative analgesia and complications in 1,416 patients. *Anesthesiology.* 2005;103(5):1035–1045.
29. Capdevila X, Bringuier S, Borgeat A. Infectious risk of continuous peripheral nerve blocks. *Anesthesiology.* 2009;110:182–188.
30. Hebl J, Niesen A. Infectious complications of regional anesthesia. *Curr Opin Anaesthesiol.* 2011;24:573–580.
31. Stafford-Smith M. Impaired haemostasis and regional anaesthesia. *Can J Anaesth.* 1996;43(5 Pt 2):R129–R141.
32. Horlocker TT, Wedel DJ, Rowlingson JC, et al. Regional anesthesia in the patient receiving antithrombotic or thrombolytic therapy: American Society of Regional Anesthesia and Pain Medicine evidence-based guidelines (third edition). *Reg Anesth Pain Med.* 2010;35(1):64–101.
33. Brull R, McCartney CJ, Chan VW, et al. Neurological complications after regional anesthesia: contemporary estimates of risk. *Anesth Analg.* 2007;104(4):965–974.
34. Borgeat A, Ekatothramis G, Kalberer F, et al. Acute and nonacute complications associated with interscalene block and shoulder surgery: a prospective study. *Anesthesiology.* 2001;95:875–880.
35. Borgeat A, Dullenkopf A, Ekatothramis G, et al. Evaluation of the lateral modified approach for continuous interscalene block after shoulder surgery. *Anesthesiology.* 2003;99:436–442.

36. Orebaugh SL, Kentor ML, Williams BA. Adverse outcomes associated with nerve stimulator-guided and ultrasound-guided peripheral nerve blocks by supervised trainees: update of a single-site database. *Reg Anesth Pain Med.* 2012;37:577–582.
37. Sites BD, Taenzer AH, Herrick MD, et al. Incidence of local anesthetic systemic toxicity and postoperative neurologic symptoms associated with 12,668 ultrasound-guided nerve blocks: an analysis from a prospective clinical registry. *Reg Anesth Pain Med.* 2012;37:478–482.
38. Welch MB, Brummett CM, Welch TD, et al. Perioperative peripheral nerve injuries: a retrospective study of 380,680 cases during a 10-year period at a single institution. *Anesthesiology.* 2009;111:490–497.
39. Barrington MJ, Kluger R. Ultrasound guidance reduces the risk of local anesthetic systemic toxicity following peripheral nerve blockade. *Reg Anesth Pain Med.* 2013;4:534–541.
40. Ecoffey C, Oger E, Marchand-Maillet F, et al. Complications associated with 27,031 ultrasound-guided axillary brachial plexus blocks: a web-based survey of 36 French centres. *Eur J Anaesthesiol.* 2014;31:606–610.
41. Neal JM, Barrington MJ, Brull R. The second ASRA practice advisory on neurologic complications associated with regional anesthesia and pain medicine executive summary 2015. *Reg Anesth Pain Med.* 2015;40:401–430.
42. Paul J, Arya A, Hurlburt L. Femoral nerve block improves analgesia outcomes after total knee arthroplasty: a meta-analysis of randomized controlled trials. *Anesthesiology.* 2010;113(5):1144–1164.
43. Richman JM, Liu SS, Courpas G, et al. Does continuous peripheral nerve block provide superior pain control to opioids? A meta-analysis. *Anesth Analg.* 2006;102(1):248–257.
44. Block BM, Liu SS, Rowlingson AJ, et al. Efficacy of postoperative epidural analgesia, a meta-analysis. *JAMA.* 2003;290(18):2455–2463.
45. Guay J. The benefits of adding epidural analgesia to general anesthesia: a meta-analysis. *J Anesth.* 2006;20:335–340.
46. Chelly JE, Greger J, Gebhard R, et al. Continuous femoral blocks improve recovery and outcome of patients undergoing total knee arthroplasty. *J Arthroplasty.* 2001;16(4):436–445.
47. Capdevila X, Barthelet Y, Biboulet P, et al. Effects of perioperative analgesic technique on the surgical outcome and duration of rehabilitation after major knee surgery. *Anesthesiology.* 1999;91(1):8–15.
48. Martin F, Martinez V, Mazoit JX, et al. Antiinflammatory effect of peripheral nerve blocks after knee surgery: clinical and biologic evaluation. *Anesthesiology.* 2008;109(3):484–490.
49. Barrington MJ, Olive D, Low K, et al. Continuous femoral nerve blockade or epidural analgesia after total knee replacement: a prospective randomized controlled trial. *Anesth Analg.* 2005;101(6):1824–1829.
50. Coley KC, Williams BA, DaPos SV, et al. Retrospective evaluation of unanticipated admissions and readmissions after same day surgery and associated costs. *J Clin Anesth.* 2002;14(5):349–353.
51. Williams BA, Kentor ML, Vogt MT, et al. Economics of nerve block pain management after anterior cruciate ligament reconstruction: potential hospital cost savings via associated postanesthesia care unit bypass and same-day discharge. *Anesthesiology.* 2004;100(3):697–706.
52. Rodgers A, Walker N, Schug S, et al. Reduction of postoperative mortality and morbidity with epidural or spinal anaesthesia: results from overview of randomized trials. *BMJ.* 2000;321(7275):1493.
53. Hu S, Zhang ZY, Hua YQ, et al. A comparison of regional and general anaesthesia for total replacement of the hip or knee: a meta-analysis. *J Bone Joint Surg Br.* 2009;91(7):935–942.
54. Chang CC, Lin HC, Lin HW, et al. Anesthetic management and surgical site infections in total hip or knee replacement: a population-based study. *Anesthesiology.* 2010;113(2):279–284.
55. Tetzlaff J. Regional anaesthetic technique and the incidence of tourniquet pain. *Can J Anaesth.* 1993;40(7):591–595.
56. Hagenouw R. Tourniquet pain: a volunteer study. *Anesth Analg.* 1986;65(11):1175–1180.
57. Oragui E. Tourniquet use in upper limb surgery. *Hand.* 2011;6:165–173.
58. Townsend H. Tourniquet release: systemic and metabolic effects. *Acta Anaesthesiol Scand.* 1996;40:1234–1237.
59. Cullen D, Kirby R. Beach chair position may decrease cerebral perfusion. Catastrophic outcomes have occurred. *APSF News.* 2007;22(2):25.
60. Pohl A, Cullen DJ. Cerebral ischemia during shoulder surgery in the upright position: a case series. *J Clin Anesth.* 2005;17:463–469.
61. Halpern S, Preston R. Postdural puncture headache and spinal needle design. Metaanalysis. *Anesthesiology.* 1994;81:1376–1383.
62. Lambert D, Hurley R, Hertwig L, et al. Role of needle gauge and tip configuration in the production of lumbar puncture headache. *Reg Anesth.* 1997;22:66–72.
63. Pittoni G, Toffoletto F, Calcarella G, et al. Spinal anesthesia in outpatient knee surgery: 22-gauge versus 25-gauge Sprotte needle. *Anesth Analg.* 1995;81:73–81.
64. Mulroy MF, Larkin KL, Hodgson PS, et al. A comparison of spinal, epidural, and general anesthesia for outpatient knee arthroscopy. *Anesth Analg.* 2000;91(4):860–864.
65. Wong J, Marshall S, Chung F, et al. Spinal anesthesia improves the early recovery profile of patients undergoing ambulatory knee arthroscopy. *Can J Anaesth.* 2001;48(4):369–374.
66. Casati A, Cappelleri G, Aldegheri G, et al. Total intravenous anesthesia, spinal anesthesia or combined sciatic-femoral nerve block for outpatient knee arthroscopy. *Minerva Anesthesiol.* 2004;70(6):493–502.
67. Korhonen AM, Valanne JV, Jokela RM, et al. A comparison of selective spinal anesthesia with hyperbaric bupivacaine and general anesthesia with desflurane for outpatient knee arthroscopy. *Anesth Analg.* 2004;99(6):1668–1673.
68. Dunn WR, Cordasco FA, Flynn E, et al. A prospective randomized comparison of spinal versus local anesthesia with propofol infusion for knee arthroscopy. *Arthroscopy.* 2006;22(5):479–483.
69. Choi S, Brull R. Is ultrasound guidance advantageous for interventional pain management? A review of acute pain outcomes. *Anesth Analg.* 2011;113(3):596–604.
70. Orebaugh SL, Williams BA, Vallejo M, et al. Adverse outcomes associated with stimulator-based peripheral nerve blocks with versus without ultrasound visualization. *Reg Anesth Pain Med.* 2009;34(3):251–255.
71. Marhofer P, Harrop-Griffiths W, Kettner SC, et al. Fifteen years of ultrasound guidance in regional anaesthesia: part 1. *Br J Anaesth.* 2010;104(5):538–546.
72. Liu SS, Ngeow J, John RS. Evidence basis for ultrasound-guided block characteristics: onset, quality, and duration. *Reg Anesth Pain Med.* 2010;35(2 suppl):S26–S35.

73. Antonakakis JG, Ting PH, Sites B. Ultrasound-guided regional anesthesia for peripheral nerve blocks: an evidence-based outcome review. *Anesthesiol Clin*. 2011;29(2):179–191.
74. Abdallah FW, Halpern SH, Aoyama K, et al. Will the real benefits of single-shot interscalene block please stand up? A systematic review and meta-analysis. *Anesth Analg*. 2015; 120(5):1114–1129.
75. Fredrickson MJ, Ball CM, Dalglish AJ. Analgesic effectiveness of a continuous versus single-injection interscalene block for minor arthroscopic shoulder surgery. *Reg Anesth Pain Med*. 2010;35(1):28–33.
76. Checucci G, Allegra A, Bigazzi P, et al. A new technique for regional anesthesia for arthroscopic shoulder surgery based on a suprascapular nerve block and an axillary nerve block: an evaluation of the first results. *Arthroscopy*. 2008;24(6):689.
77. Auyong DB, Yuan SC, Choi DS, et al. A double-blind randomized comparison of continuous interscalene, supraclavicular, and suprascapular blocks for total shoulder arthroplasty. *Reg Anesth Pain Med*. 2017;42(3):302309.
78. Brown A. Anaesthesia for procedures of the hand and elbow. *Best Pract Res Clin Anaesthesiol*. 2002;16(2):227–246.
79. Heil JW, Ilfeld BM, Loland VJ, et al. Preliminary experience with a novel ultrasound-guided supraclavicular perineural catheter insertion technique for perioperative analgesia of the upper extremity. *J Ultrasound Med*. 2010;29(10):1481–1485.
80. Kim BG, Han JU, Song JH, et al. A comparison of ultrasound-guided interscalene and supraclavicular blocks for post-operative analgesia after shoulder surgery. *Acta Anaesthesiol Scand*. 2017;61(4):427–435.
81. Ward JP, Albert DB, Altman R, et al. Are femoral nerve blocks effective for early postoperative pain management after hip arthroscopy? *Arthroscopy*. 2012;28:1064–1069.
82. Zhang Z, Zhu W, Zhu L, et al. Efficacy of celecoxib for pain management after arthroscopic surgery of hip: a prospective randomized placebo-controlled study. *Eur J Orthop Surg Traumatol*. 2014;24:919–923.
83. Morgenthaler K, Bauer C, Ziegler S, et al. Intra-articular bupivacaine following hip joint arthroscopy. Effect on postoperative pain. *Anaesthesist*. 2007;56:1128–1132.
84. Baker JF, McGuire CM, Byrne DP, et al. Analgesic control after hip arthroscopy: a randomized, double-blinded trial comparing portal with intra-articular infiltration of bupivacaine. *Hip Int*. 2011;21:373–377.
85. Baker JF, Byrne DP, Hunter K, et al. Post-operative opiate requirements after hip arthroscopy. *Knee Surg Sports Traumatol Arthrosc*. 2011;19:1399–1402.
86. Bech NH, Hulst AH, Spuijbroek JA, et al. Perioperative pain management in hip arthroscopy; what options are there? *J Hip Preserv Surg*. 2016;3(3):181–189.
87. Potter MQ, Sun GS, Fraser JA, et al. Psychological distress in hip arthroscopy patients affects postoperative pain control. *Arthroscopy*. 2014;30:195–201.
88. Tran KM, Ganley TJ, Wells L, et al. Intraarticular bupivacaine-clonidine-morphine versus femoral-sciatic nerve block in pediatric patients undergoing anterior cruciate ligament reconstruction. *Anesth Analg*. 2005;101:1304–1310.
89. Dauri M, Polzoni M, Fabbi E, et al. Comparison of epidural, continuous femoral block and intraarticular analgesia after anterior cruciate ligament reconstruction. *Acta Anaesthesiol Scand*. 2003;47:20–25.
90. Williams BA, Kentor ML, Vogt MT, et al. Femoral-sciatic nerve blocks for complex outpatient knee surgery are associated with less postoperative pain before same-day discharge: a review of 1,200 consecutive cases from the period 1996–1999. *Anesthesiology*. 2003;98:1206–1213.
91. Harbell MW, Cohen JM, Kolodzie K, et al. Combined preoperative femoral and sciatic nerve blockade improves analgesia after anterior cruciate ligament reconstruction: a randomized controlled clinical trial. *J Clin Anesth*. 2016;33: 68–74.
92. Williams BA, Kentor ML, Vogt MT, et al. Reduction of verbal pain scores after anterior cruciate ligament reconstruction with 2-day continuous femoral nerve block: a randomized clinical trial. *Anesthesiology*. 2006;104(2):315–327.
93. Hussain N, Ferreri TG, Prusick PJ, et al. Adductor canal block versus femoral canal block for total knee arthroplasty: a meta-analysis: what does the evidence suggest? *Reg Anesth Pain Med*. 2016;41:314–320.
94. Abdallah FW, Whelan DB, Chan VW, et al. Adductor canal block provides noninferior analgesia and superior quadriceps strength compared with femoral nerve block in anterior cruciate ligament reconstruction. *Anesthesiology*. 2016;124: 1053–1064.
95. di Benedetto P, Casati A, Bertini L, et al. Postoperative analgesia with continuous sciatic nerve block after foot surgery: a prospective, randomized comparison between the popliteal and subgluteal approaches. *Anesth Analg*. 2002;94(4):996–1000.
96. Ilfeld BM, Madison SJ. The sciatic nerve and knee arthroplasty: to block, or not to block—that is the question. *Reg Anesth Pain Med*. 2011;36(5):421–423.
97. Benzon H, Shara S, Calimaran A. Comparison of different approaches to the saphenous nerve block. *Anesthesiology*. 2005;102:633–638.
98. Tucker G, Boas R. Pharmacokinetic aspects of intravenous regional anesthesia. *Anesthesiology*. 1971;34:538.
99. Hall MJ, Lawrence L. Ambulatory surgery in the United States. 1996. *Advances Data* 1–16. 1998.
100. Burks RT, Tashjian RZ, Swenson J, et al. Diaphragm function and room air oxygen saturation are affected less with 0.125% bupivacaine interscalene blocks. Paper presented at: 77th Annual Meeting of the American Academy of Orthopaedic Surgeons. New Orleans, LA. Abstract P291. 2010.
101. Swenson JD, Bay N, Loose E, et al. Outpatient management of continuous peripheral nerve catheters placed using ultrasound guidance: an experience in 620 patients. *Anesth Analg*. 2006;103(6):1436–1443.
102. Berde C. Local anesthetics. In: Miller R, ed. *Miller's Anesthesia*. Elsevier; 2009:913–939.
103. Strichartz G. Fundamental properties of local anesthetics. II. Measured octanol: buffer partition coefficients and pKa values of clinically used drugs. *Anesth Analg*. 1990;71(2):158–170.
104. Klein S. A comparison of 0.5% bupivacaine, 0.5% ropivacaine, and 0.75% ropivacaine for interscalene brachial plexus block. *Reg Anesth Pain Med*. 1998;17:1316–1319.
105. Liu S. Epinephrine prolongs duration of subcutaneous infiltration of local anesthesia in a dose-related manner. Correlation with magnitude of vasoconstriction. *Reg Anesth*. 1995;20(5):378–384.
106. Di Gregorio G. Clinical presentation of local anesthetic systemic toxicity. *Reg Anesth Pain Med*. 2010;35(2):181–187.
107. Burney R. Effects of pH on protein binding of lidocaine. *Anesth Analg*. 1978;57(4):478–480.
108. Butterworth J. Models and mechanisms of local anesthetic cardiac toxicity. *Reg Anesth Pain Med*. 2010;35(2):167–176.

109. Clarkson C. Mechanism for bupivacaine depression of cardiac conduction: fast block of sodium channels during the action potential with slow recovery from block during diastole. *Anesthesiology*. 1985;62:396–405.
110. Neal JM, Bernards CM, Butterworth JF, et al. ASRA Practice advisory on local anesthetic systemic toxicity. *Reg Anesth Pain Med*. 2010;35(2):151–161.
111. Wolfe J. Local anesthetic systemic toxicity: update on mechanisms and treatment. *Curr Opin Anaesthesiol*. 2011;24(5):561–566.
112. <https://www.cdc.gov/drugoverdose/epidemic/Understanding-the-epidemic-drug-overdose>. Centers for Disease Control and Prevention website. <http://www.cdc.gov/drugoverdose/epidemic/>. Accessed September 12, 2017.
113. Sun EC, Darnall BD, Baker LC, et al. Incidence of and risk factors for chronic opioid use among opioid-naïve patients in the postoperative period. *JAMA Intern Med*. 2016;176(9):1286–1293.
114. Kim N, Matzon JL, Abboudi J, et al. A prospective evaluation of opioid utilization after upper-extremity surgical procedures: identifying consumption patterns and determining prescribing guidelines. *J Bone Joint Surg Am*. 2016;98(20):pe89.
115. Waljee JF, Zhong L, Hou H, et al. The use of opioid analgesics following common upper extremity surgical procedures: a national, population-based study. *Plast Reconstr Surg*. 2016;137(2):355e–364e.
116. Chou R, Gordon DB, de Leon-Casasola OA, et al. Management of postoperative pain: a clinical practice guideline from the American Pain Society, the American Society of Regional Anesthesia and Pain Medicine and the American Society of Anesthesiologists' Committee on Regional Anesthesia, executive committee, and administrative council. *J Pain*. 2016;17(2):131–157.
117. Kehlet H. Multimodal approach to control postoperative pathophysiology and rehabilitation. *Br J Anaesth*. 1997;78:606–617.
118. Sarhill N. Hydromorphone: pharmacology and clinical applications in cancer patients. *Support Care Cancer*. 2001;9(2):84–96.
119. Gottschalk A, Durieux ME, Nemergut EC. Intraoperative methadone improves postoperative pain control in patients undergoing complex spine surgery. *Anesth Analg*. 2011;112(1):218–223.
120. Gourlay GK, Wilson PR, Glynn CJ. Pharmacodynamics and pharmacokinetics of methadone during the perioperative period. *Anesthesiology*. 1982;57:458–467.
121. Gourlay GK, Willis RJ, Wilson PR. Postoperative pain control with methadone: influence of supplementary methadone doses and blood concentration-response relationships. *Anesthesiology*. 1984;61:19–26.
122. Gourlay GK, Willis RJ, Lamberty J. A double-blind comparison of the efficacy of methadone and morphine in postoperative pain control. *Anesthesiology*. 1986;64:322–327.
123. Anderson TA, Quaye ANA, Ward EN, et al. To stop or not stop, that is the question. Acute pain management for the patient on chronic buprenorphine. *Anesthesiology*. 2017;126(6):1180–1186.
124. Oscier C, Milner Q. Perioperative use of paracetamol. *Anaesthesia*. 2009;64:65–72.
125. Macario A. A literature review of randomized clinical trials of intravenous acetaminophen (paracetamol) for acute postoperative pain. *Pain Pract*. 2010;11(3):290–296.
126. Sinatra RS, Jahr JS, Reynolds L, et al. Intravenous acetaminophen for pain after major orthopedic surgery: an expanded analysis. *Pain Pract*. 2012;12(5):357–365.
127. Coloma M. The effect of ketorolac on recovery after anorectal surgery: intravenous versus local administration. *Anesth Analg*. 2000;90(5):1107–1110.
128. De Oliveira G. Perioperative single dose ketorolac to prevent postoperative pain: a meta-analysis of randomized trials. *Anesth Analg*. 2012;114(2):424–433.
129. Kohrs R. Ketamine: teaching an old drug new tricks. *Anesth Analg*. 1998;87(5):1186–1193.
130. Menigaux C. The benefits of intraoperative small-dose ketamine on postoperative pain after anterior cruciate ligament repair. *Anesth Analg*. 2000;90(1):129–135.
131. Suzuki M. Small-dose ketamine enhances morphine-induced analgesia after outpatient surgery. *Anesth Analg*. 1999;89(1):98–103.
132. Nelson LE, Lu J, Guo T, et al. The alpha2-adrenoceptor agonist dexmedetomidine converges on an endogenous sleep-promoting pathway to exert its sedative effects. *Anesthesiology*. 2003;98(2):428–436.
133. Gerlach AT, Dasta JF. Dexmedetomidine: an updated review. *Ann Pharmacother*. 2007;41(2):245–252.
134. Practice guidelines for sedation and analgesia by non-anesthesiologists: an updated report by the American Society of Anesthesiologists Task Force on Sedation and Analgesia by Non-Anesthesiologists. *Anesthesiology*. 2005;96:1004–1017.
135. Dewachter P. What investigation after an anaphylactic reaction during anaesthesia? *Curr Opin Anaesthesiol*. 2008;21(3):363–368.
136. Gurrieri C. Allergic reactions during anesthesia at a large United States referral center. *Anesth Analg*. 2011;113(5):1202–1212.
137. Dewachter P. Anaphylaxis and anesthesia: controversies and new insights. *Anesthesiology*. 2009;111(5):1141–1150.
138. Beghi E, Kurland LT, Mulder DW, et al. Brachial plexus neuropathy in the population of Rochester, Minnesota. *Ann Neurol*. 1985;18:320–323.
139. Feinburg J. Fall von Erb-Klumpke scher Lahmung nach influenza. *Centralbl*. 1897;16:588–637.
140. Parsonage M, Turner J. Neuralgic amyotrophy: the shoulder-girdle syndrome. *Lancet*. 1948;1:973–978.
141. Tsairis P, Dyck PJ, Mulder DW. Natural history of brachial plexus neuropathy. Report of 99 patients. *Arch Neurol*. 1972;27:109–127.
142. Misamore GW, Lehman DE. Parsonage-Turner syndrome (acute brachial neuritis). *J Bone Joint Surg*. 1996;78A:1405–1408.